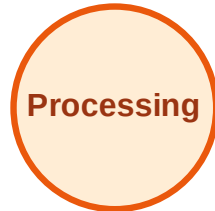


BFS Traversal

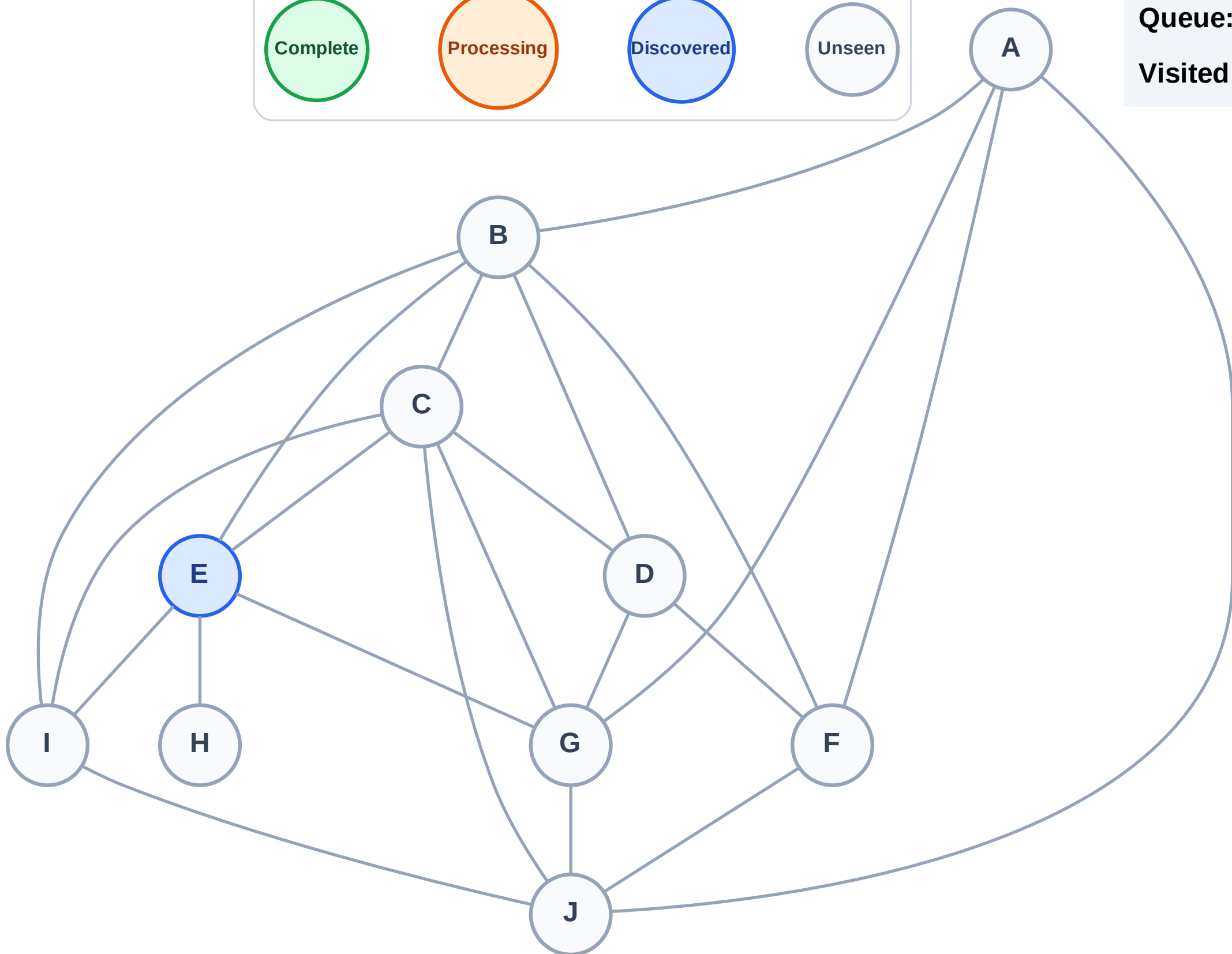
Step 1: Start at E

Legend



Queue: E

Visited: (none)



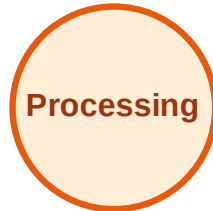
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



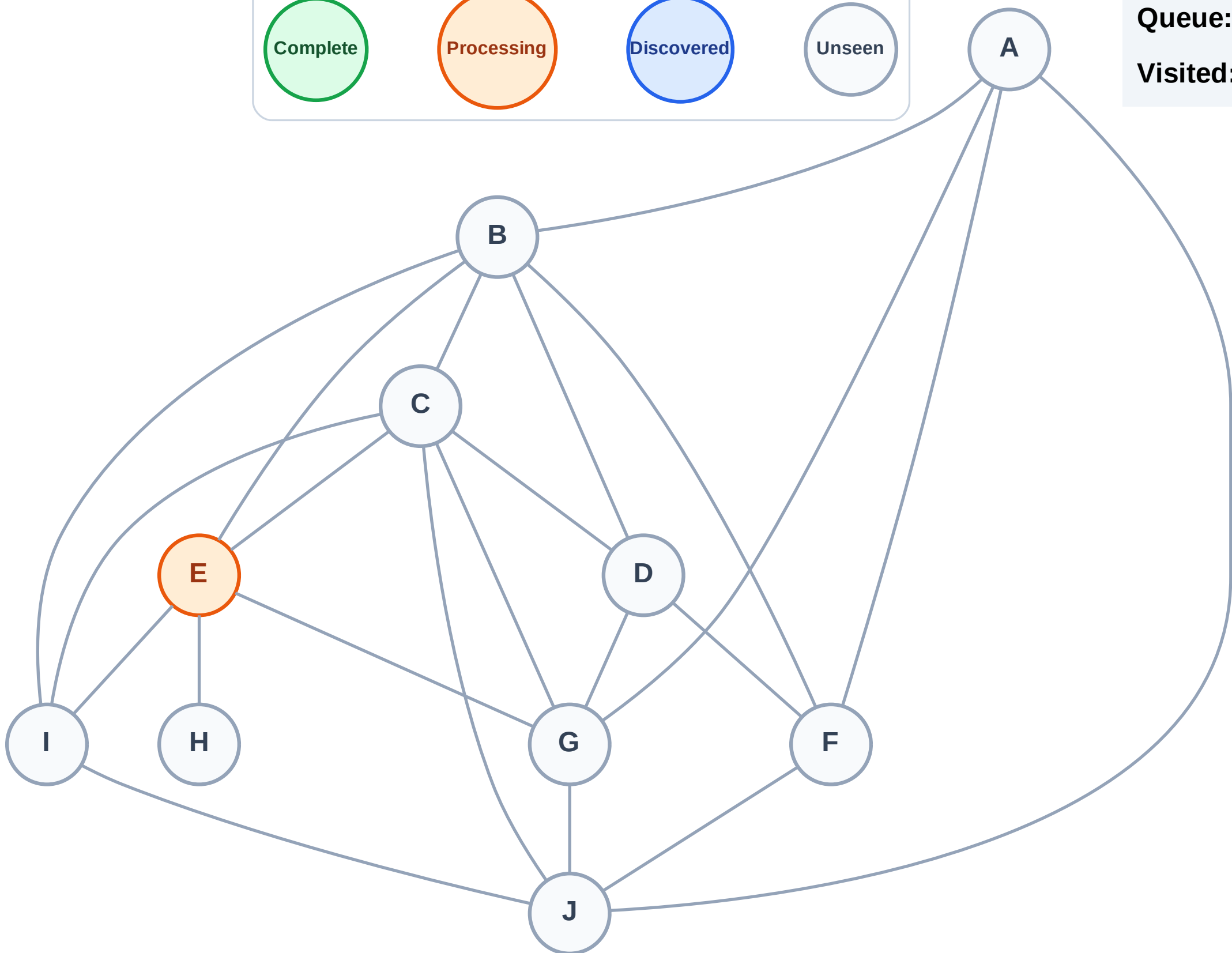
Discovered



Unseen

Queue: (empty)

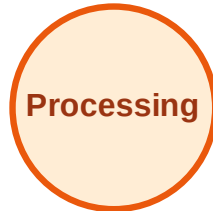
Visited: (none)



BFS Traversal

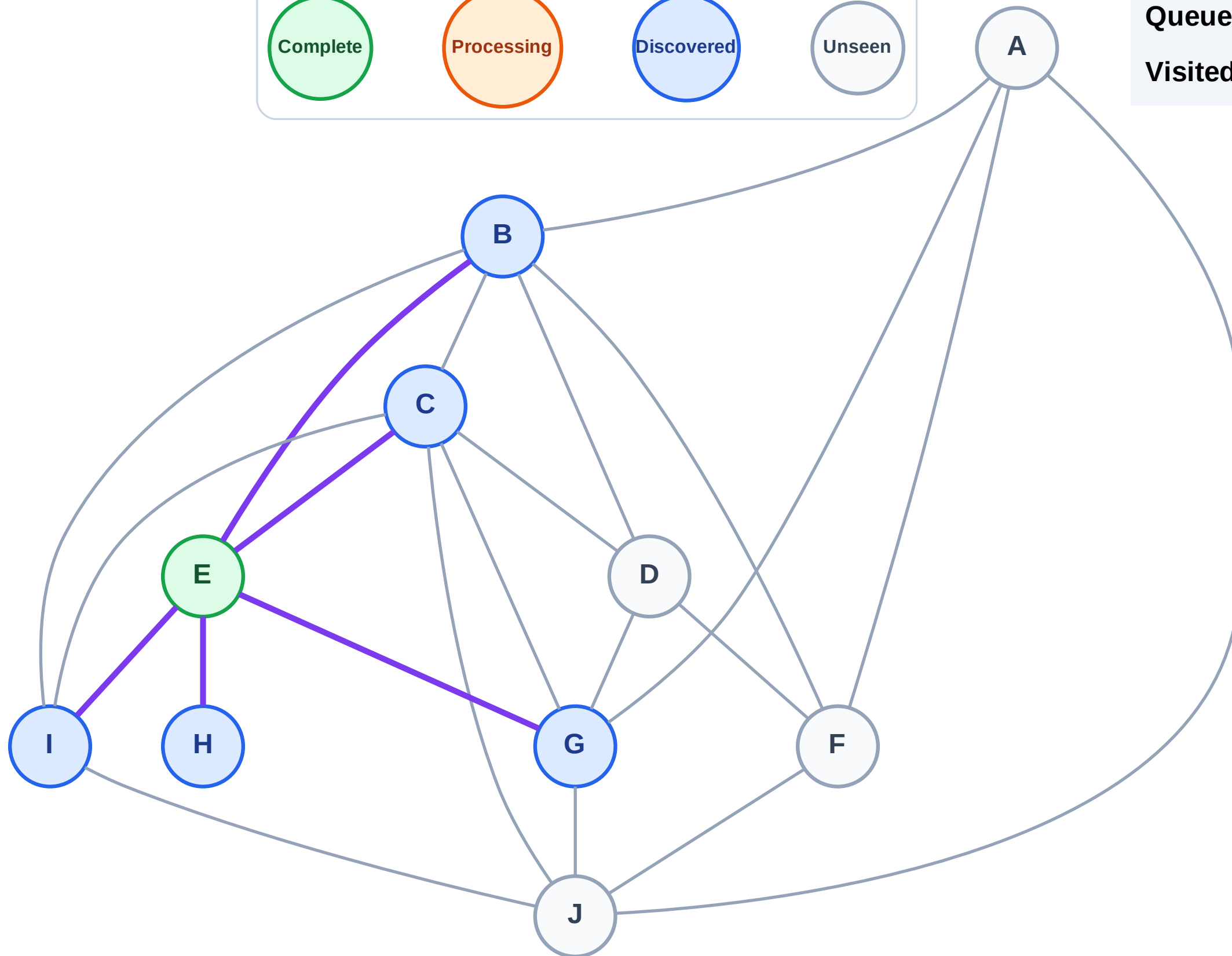
Step 3: Discover B, C, G, H, I from E

Legend



Queue: B → C → G → H → I

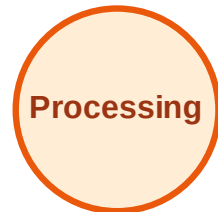
Visited: E



BFS Traversal

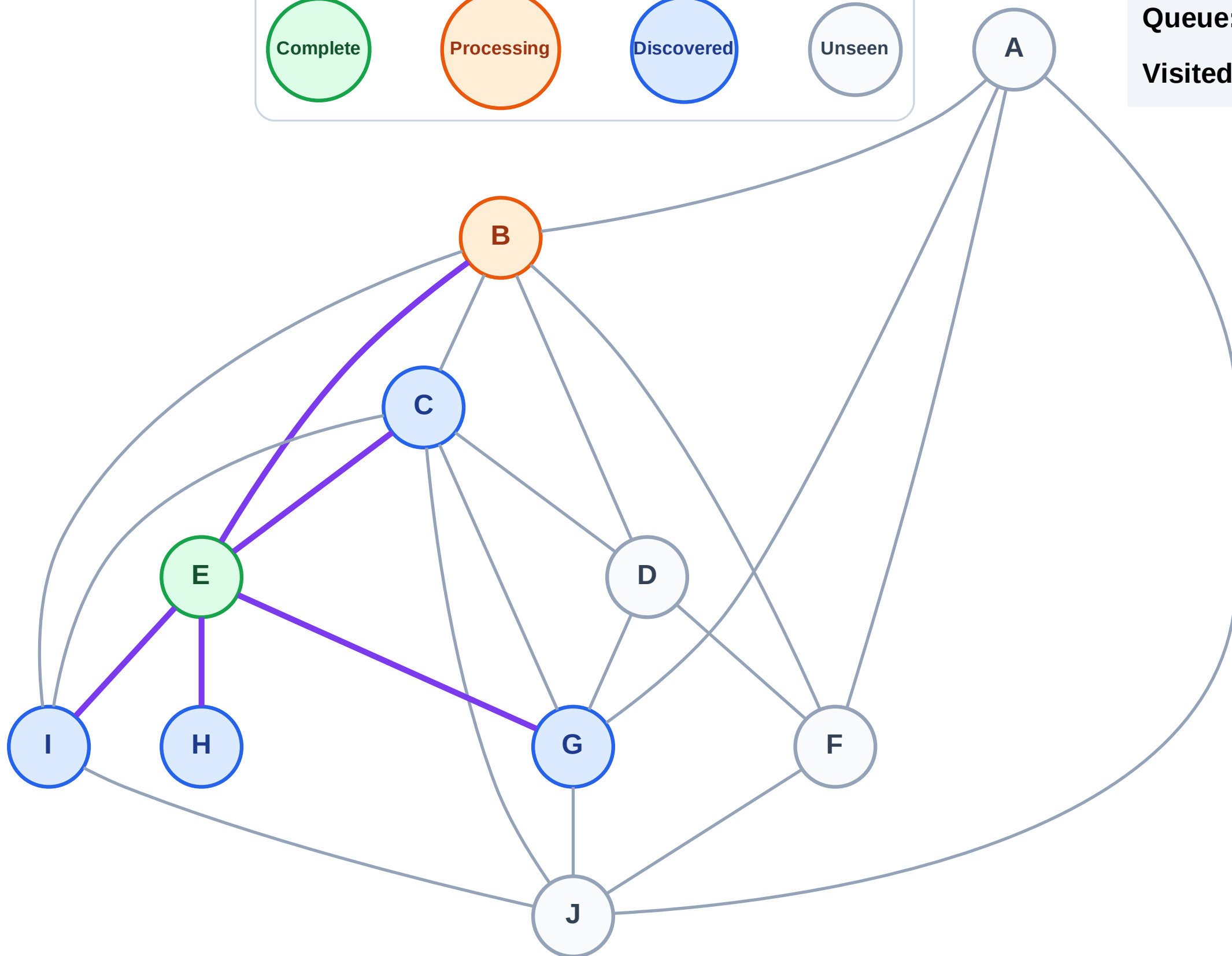
Step 4: Process node B

Legend



Queue: C → G → H → I

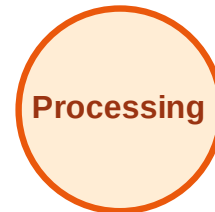
Visited: E



BFS Traversal

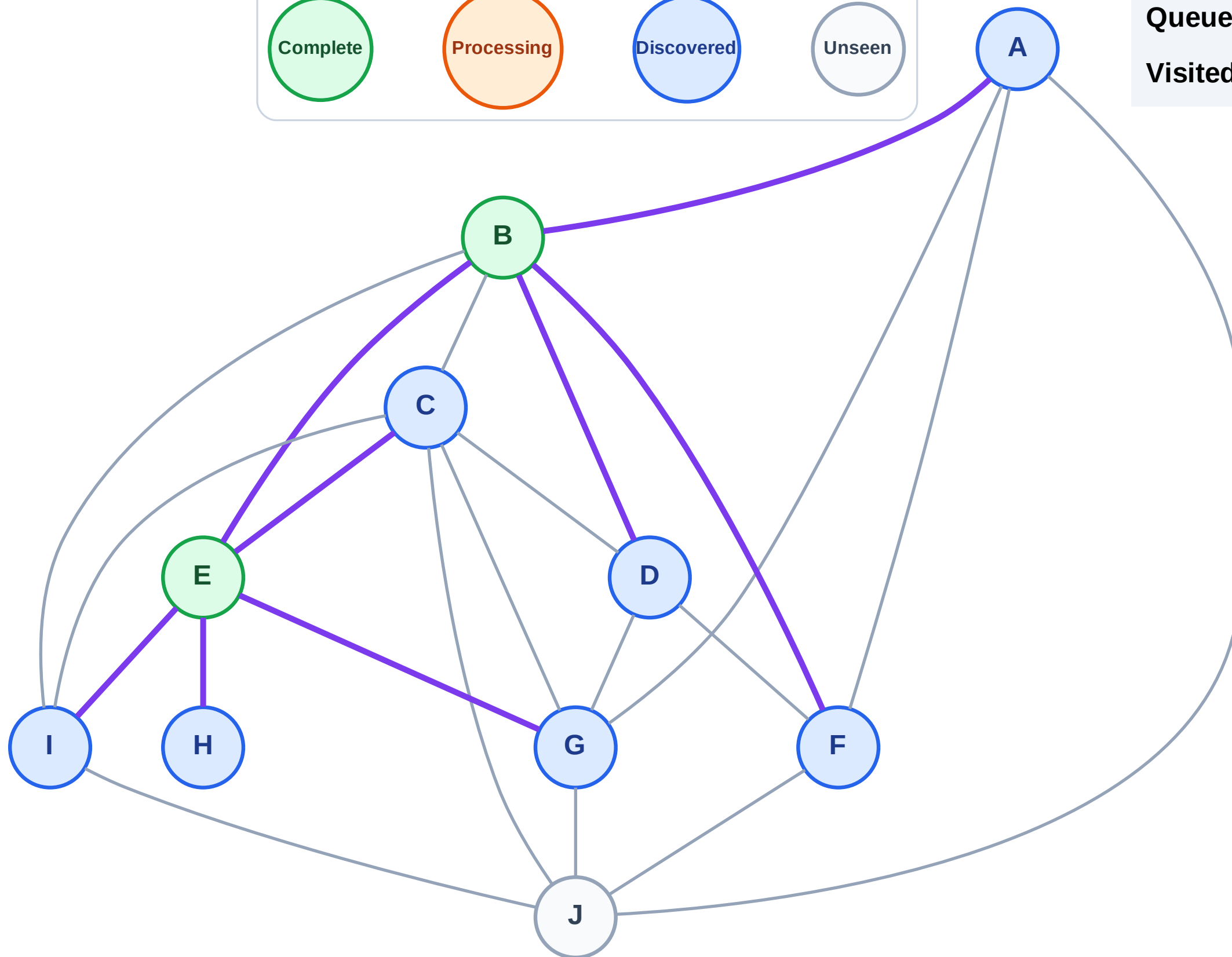
Step 5: Discover A, D, F from B

Legend



Queue: C → G → H → I → A → D → F

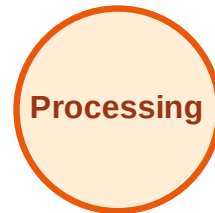
Visited: B, E



BFS Traversal

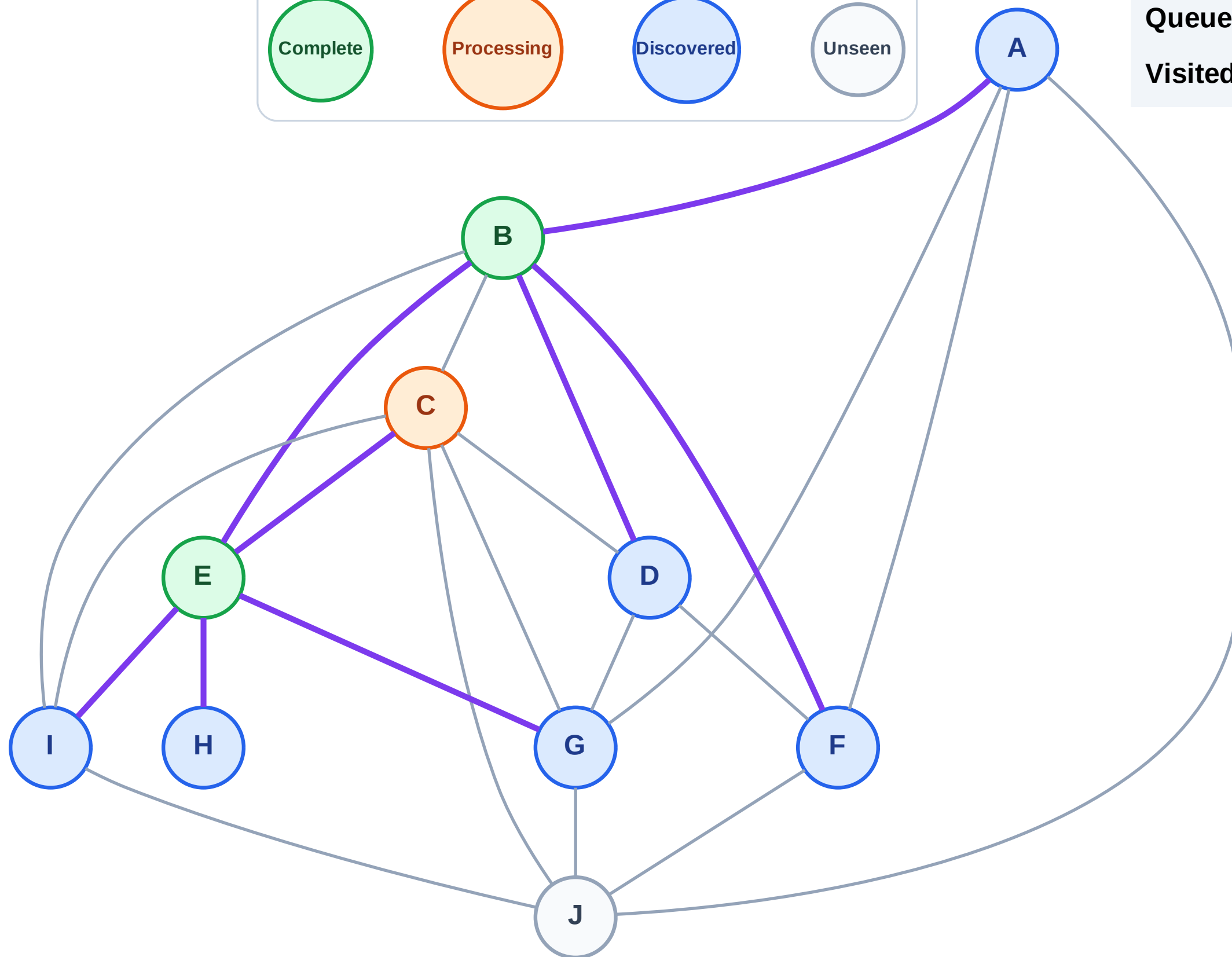
Step 6: Process node C

Legend



Queue: G → H → I → A → D → F

Visited: B, E



BFS Traversal

Step 7: Discover J from C

Legend

Complete

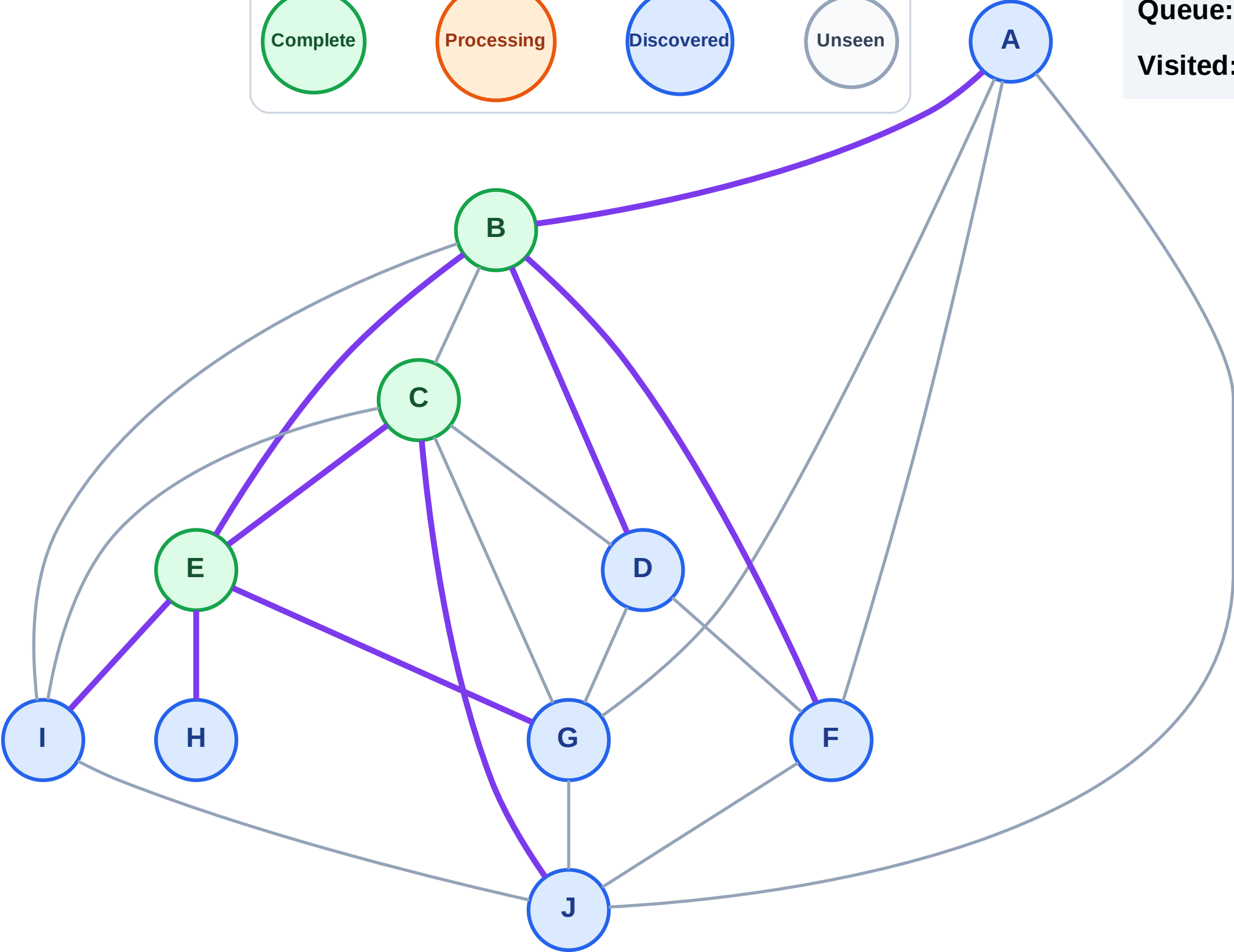
Processing

Discovered

Unseen

Queue: G → H → I → A → D → F → J

Visited: B, C, E



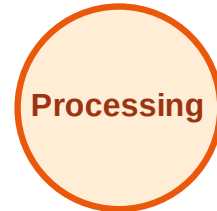
BFS Traversal

Step 8: Process node G

Legend



Complete



Processing



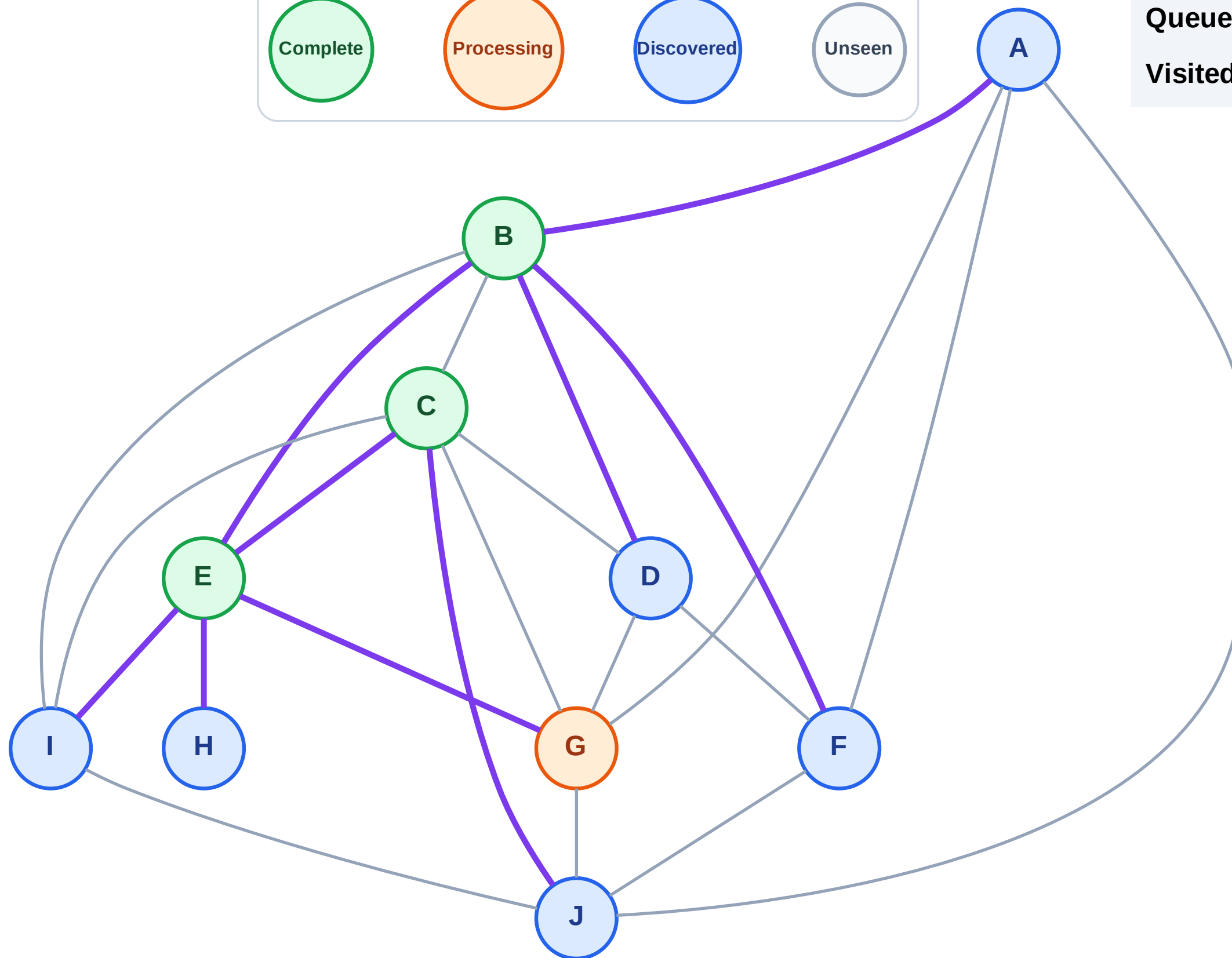
Discovered



Unseen

Queue: H → I → A → D → F → J

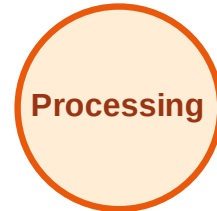
Visited: B, C, E



BFS Traversal

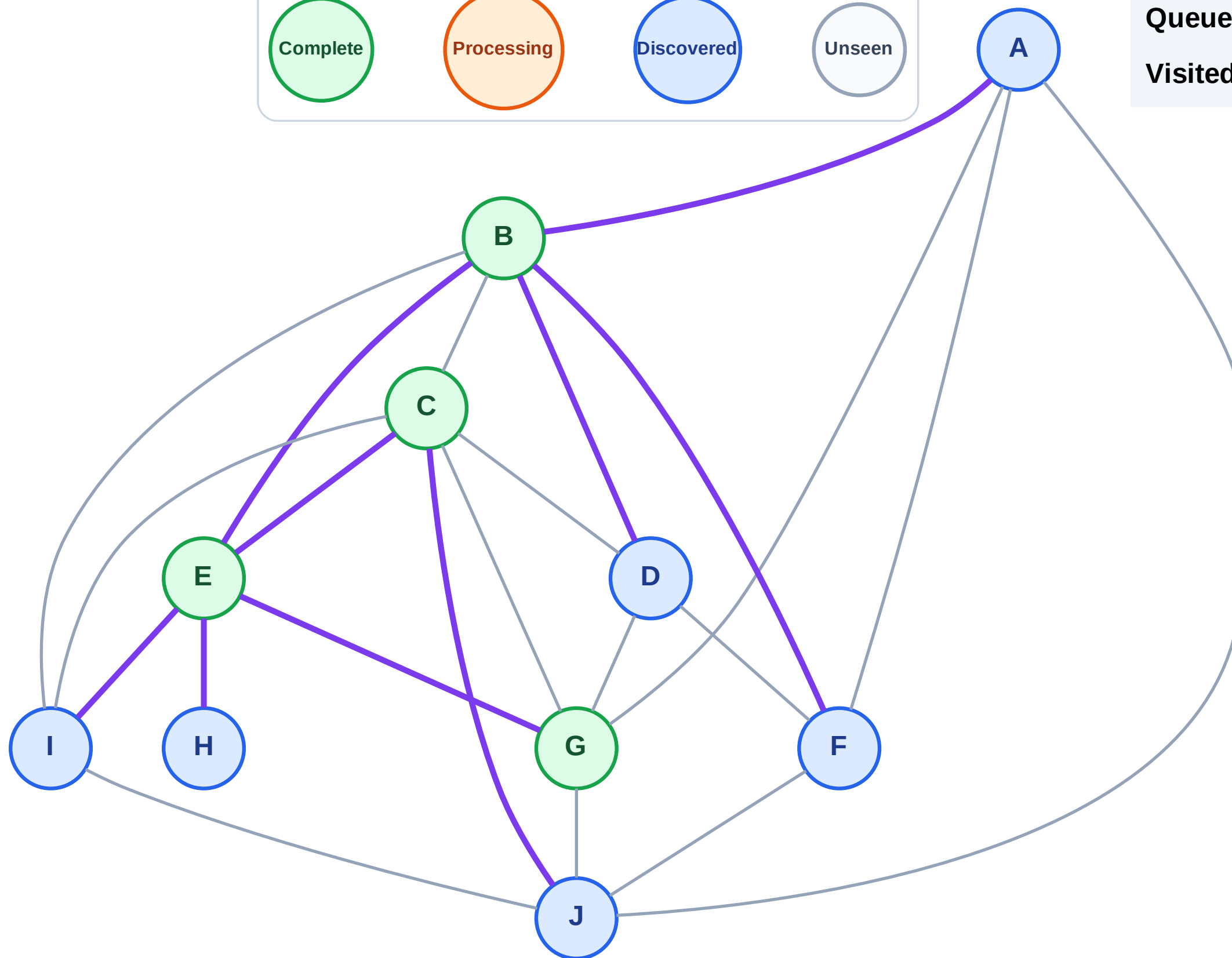
Step 9: No new neighbors from G

Legend



Queue: H → I → A → D → F → J

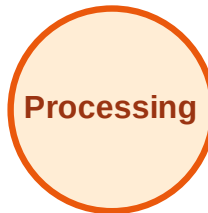
Visited: B, C, E, G



BFS Traversal

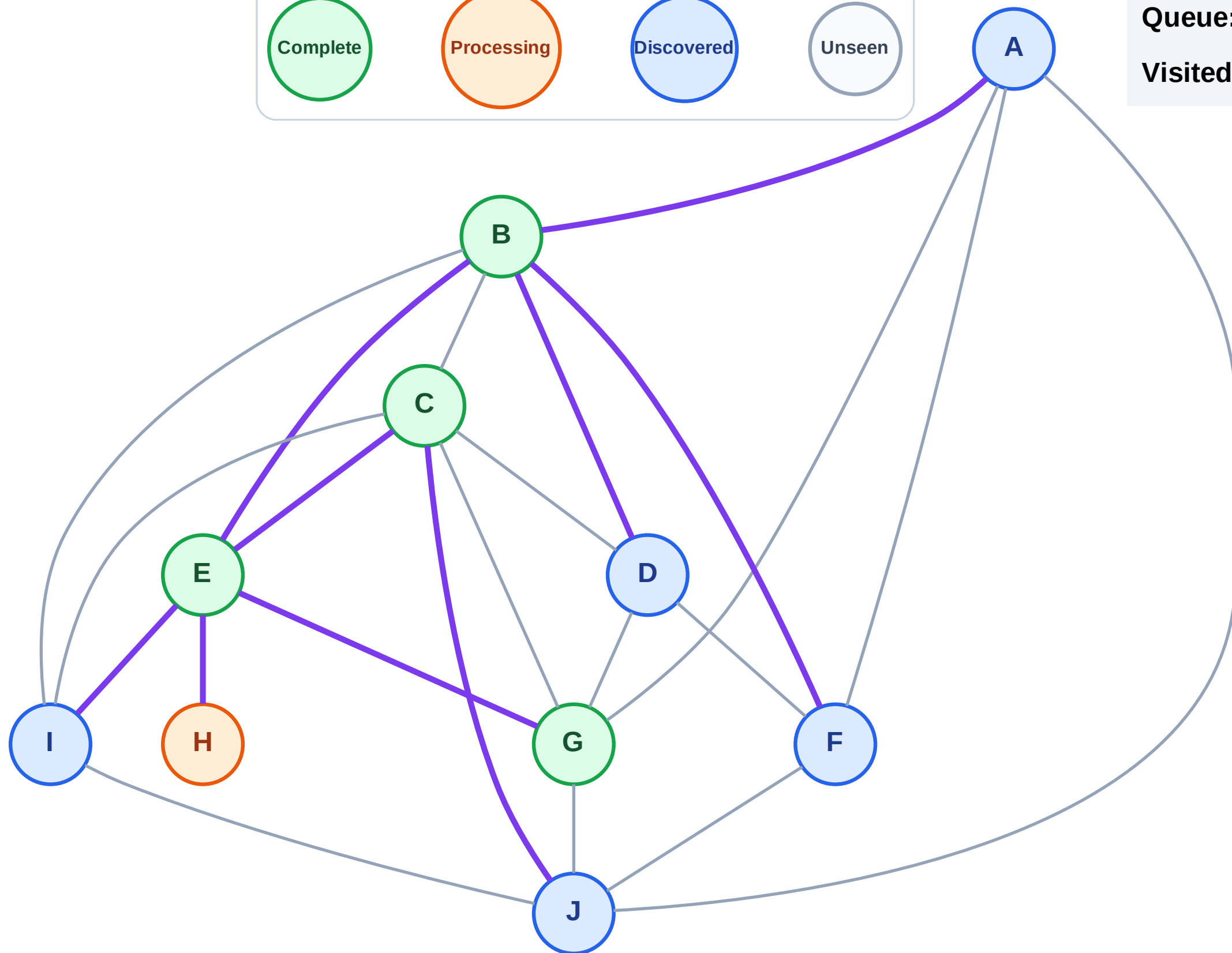
Step 10: Process node H

Legend



Queue: I → A → D → F → J

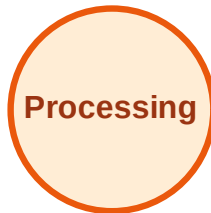
Visited: B, C, E, G



BFS Traversal

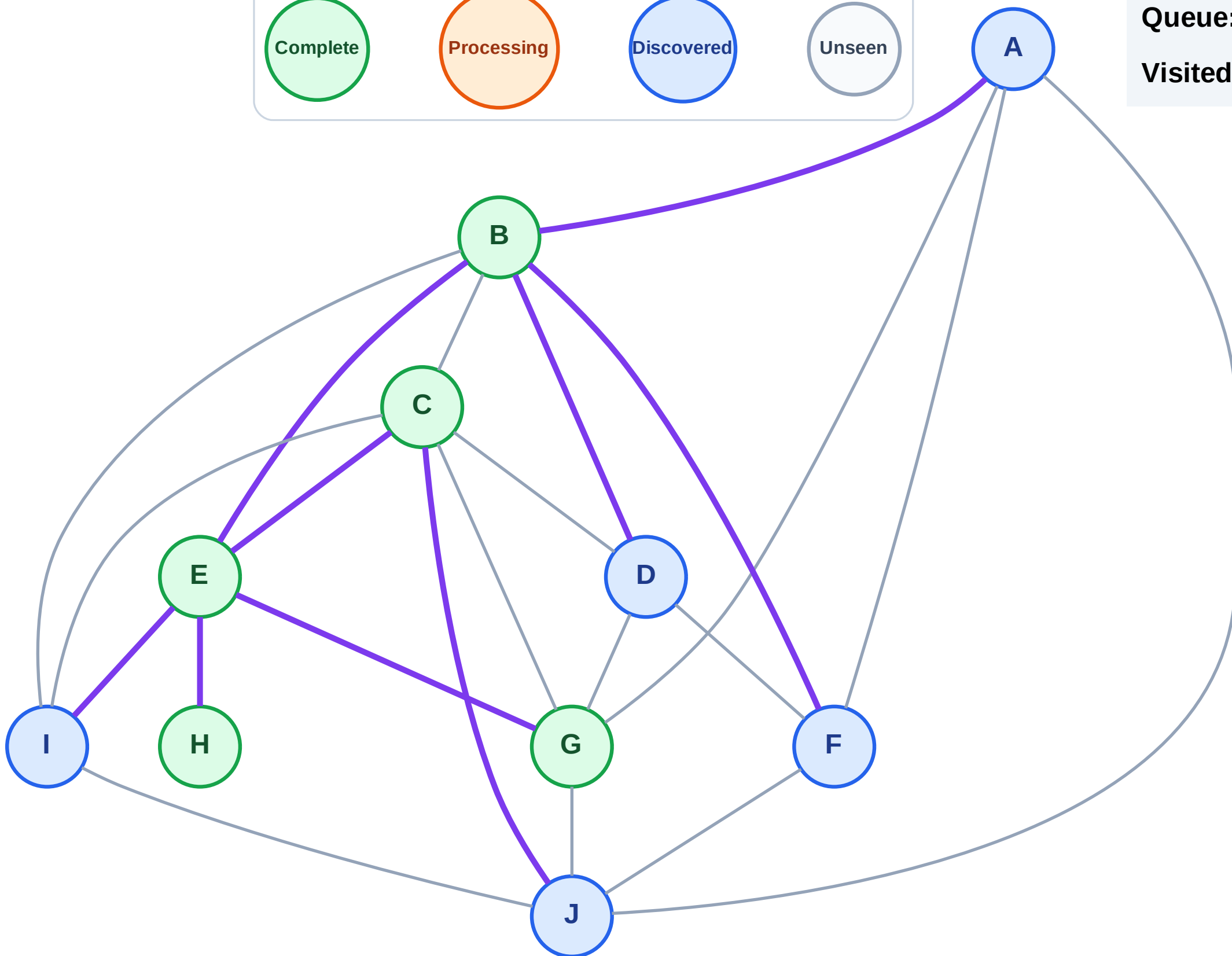
Step 11: No new neighbors from H

Legend



Queue: I → A → D → F → J

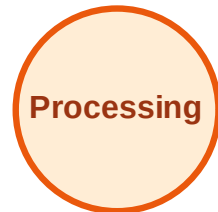
Visited: B, C, E, G, H



BFS Traversal

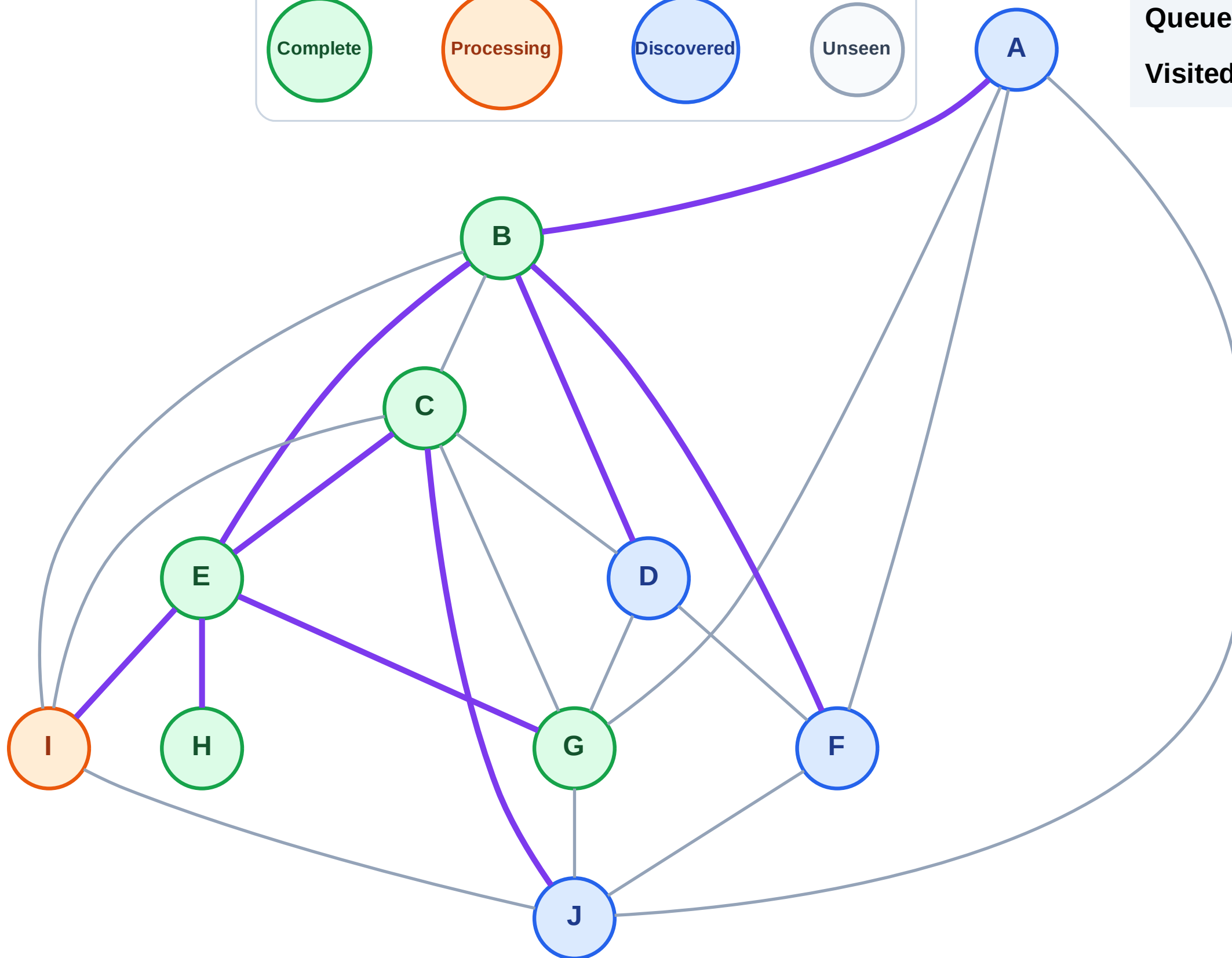
Step 12: Process node I

Legend



Queue: A → D → F → J

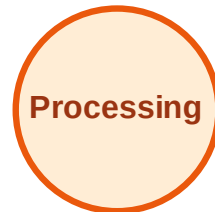
Visited: B, C, E, G, H



BFS Traversal

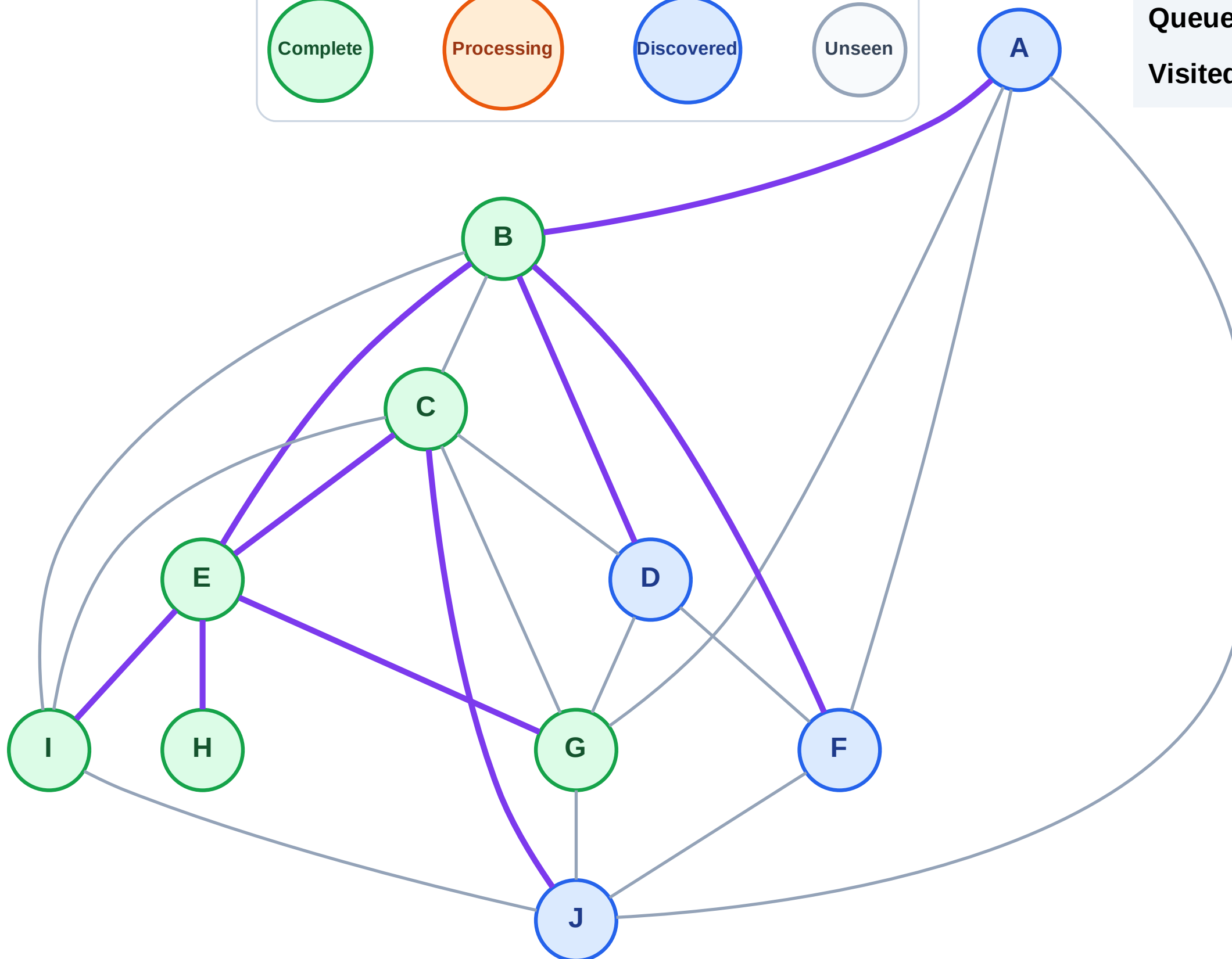
Step 13: No new neighbors from I

Legend



Queue: A → D → F → J

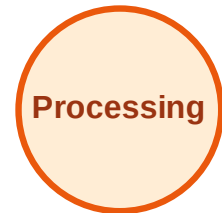
Visited: B, C, E, G, H, I



BFS Traversal

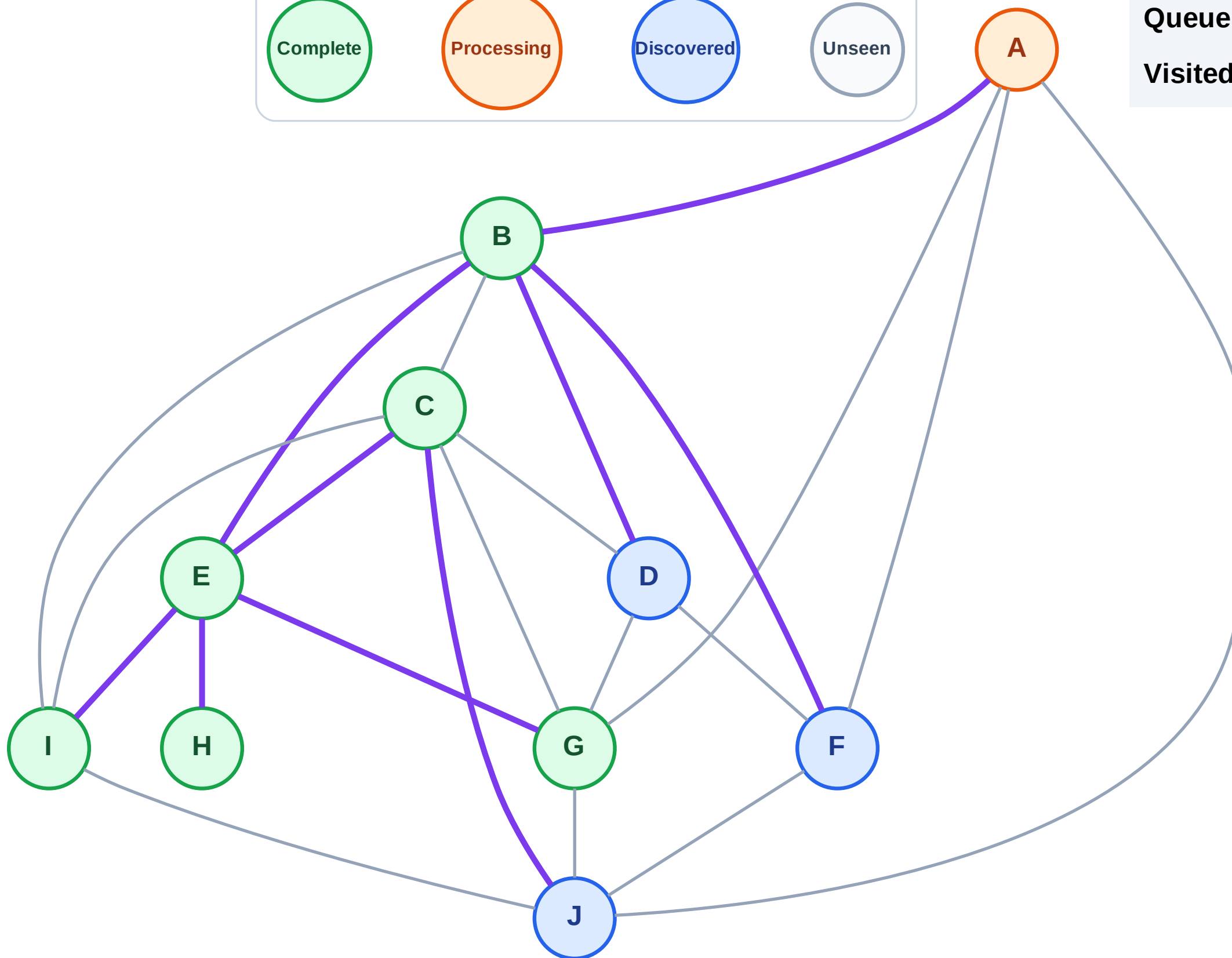
Step 14: Process node A

Legend



Queue: D → F → J

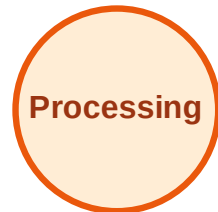
Visited: B, C, E, G, H, I



BFS Traversal

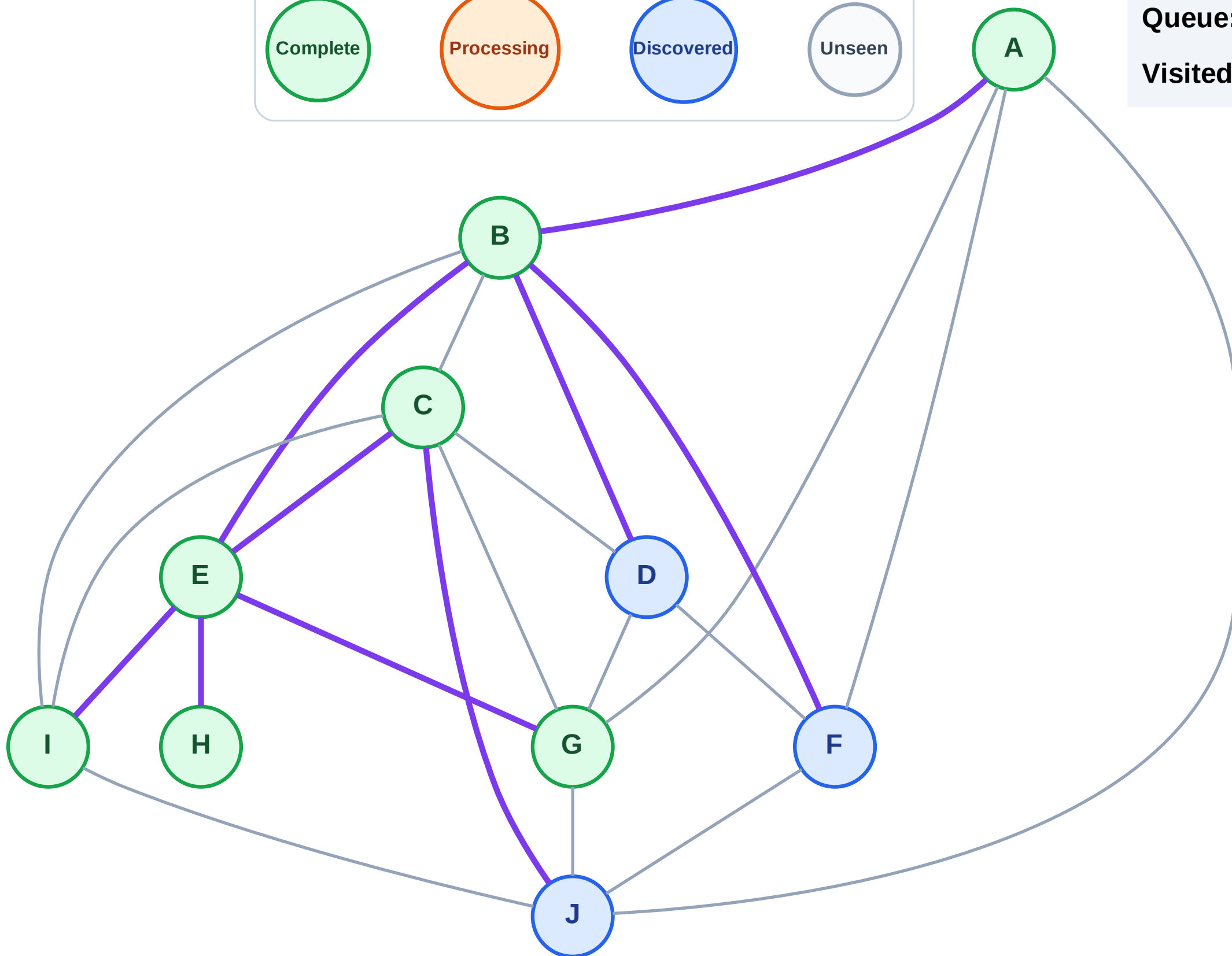
Step 15: No new neighbors from A

Legend



Queue: D → F → J

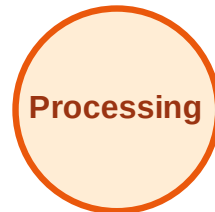
Visited: A, B, C, E, G, H, I



BFS Traversal

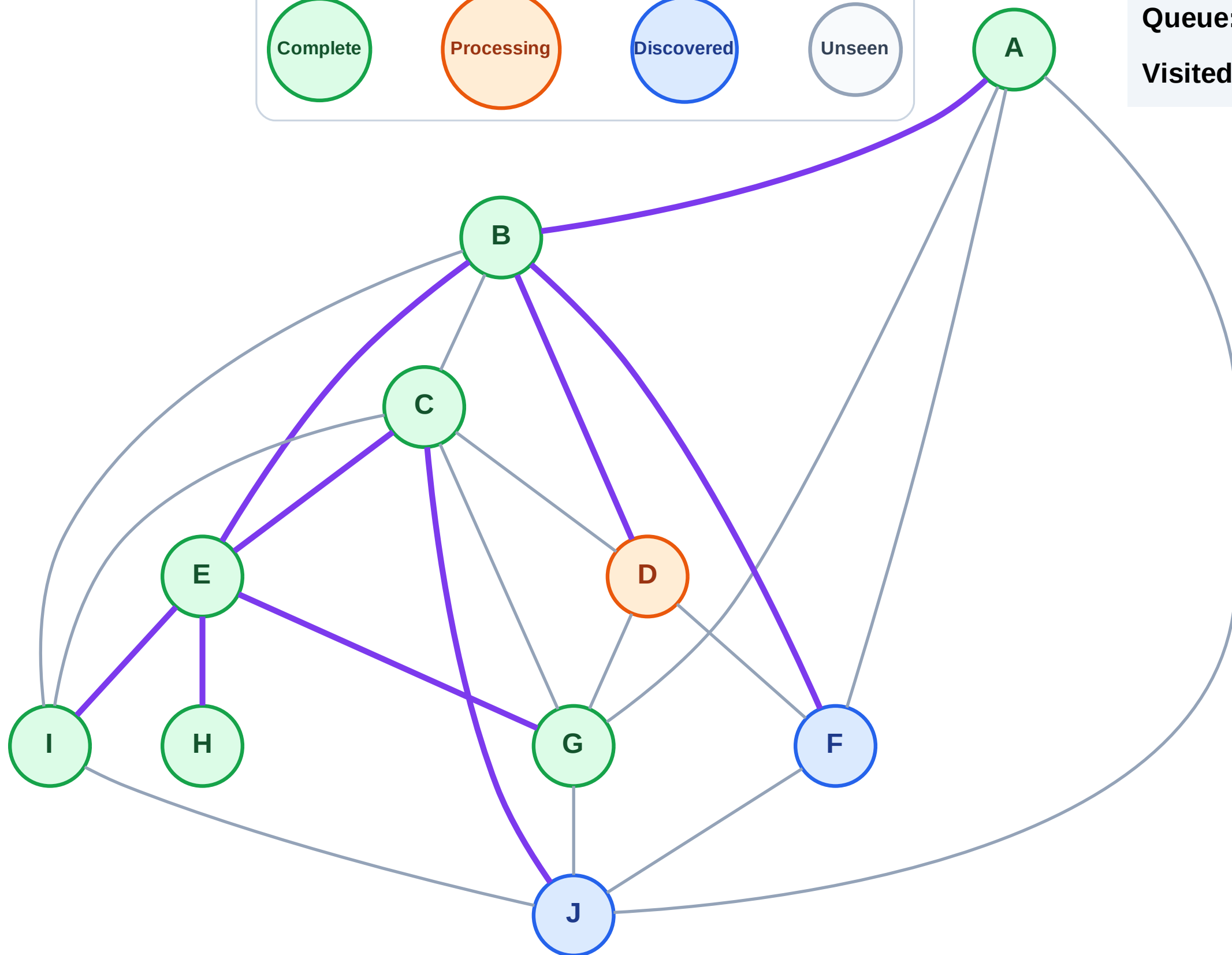
Step 16: Process node D

Legend



Queue: F → J

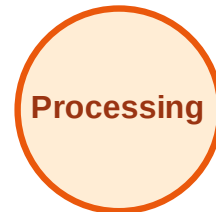
Visited: A, B, C, E, G, H, I



BFS Traversal

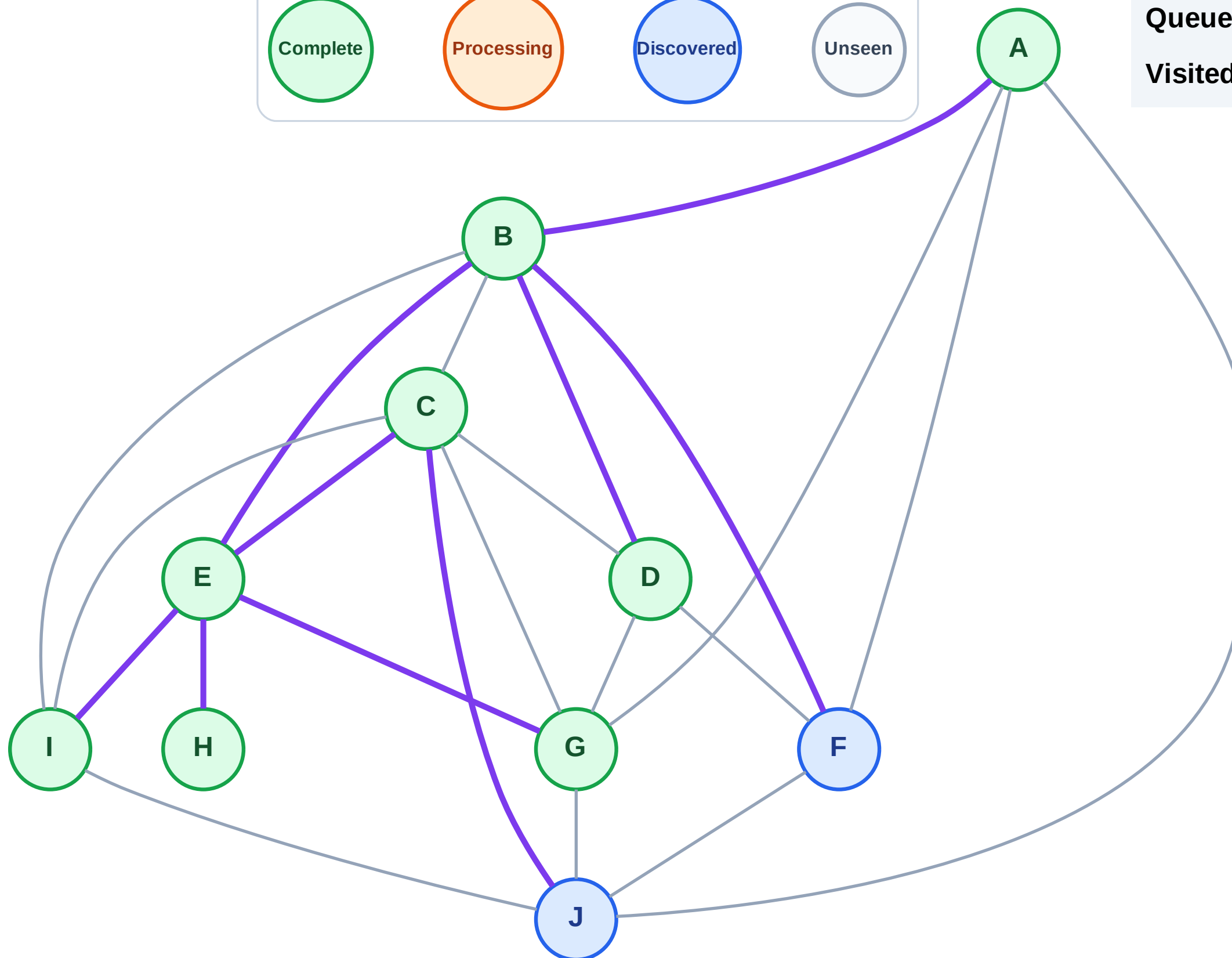
Step 17: No new neighbors from D

Legend



Queue: F → J

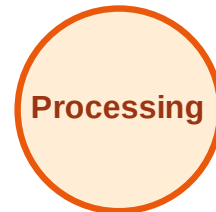
Visited: A, B, C, D, E, G, H, I



BFS Traversal

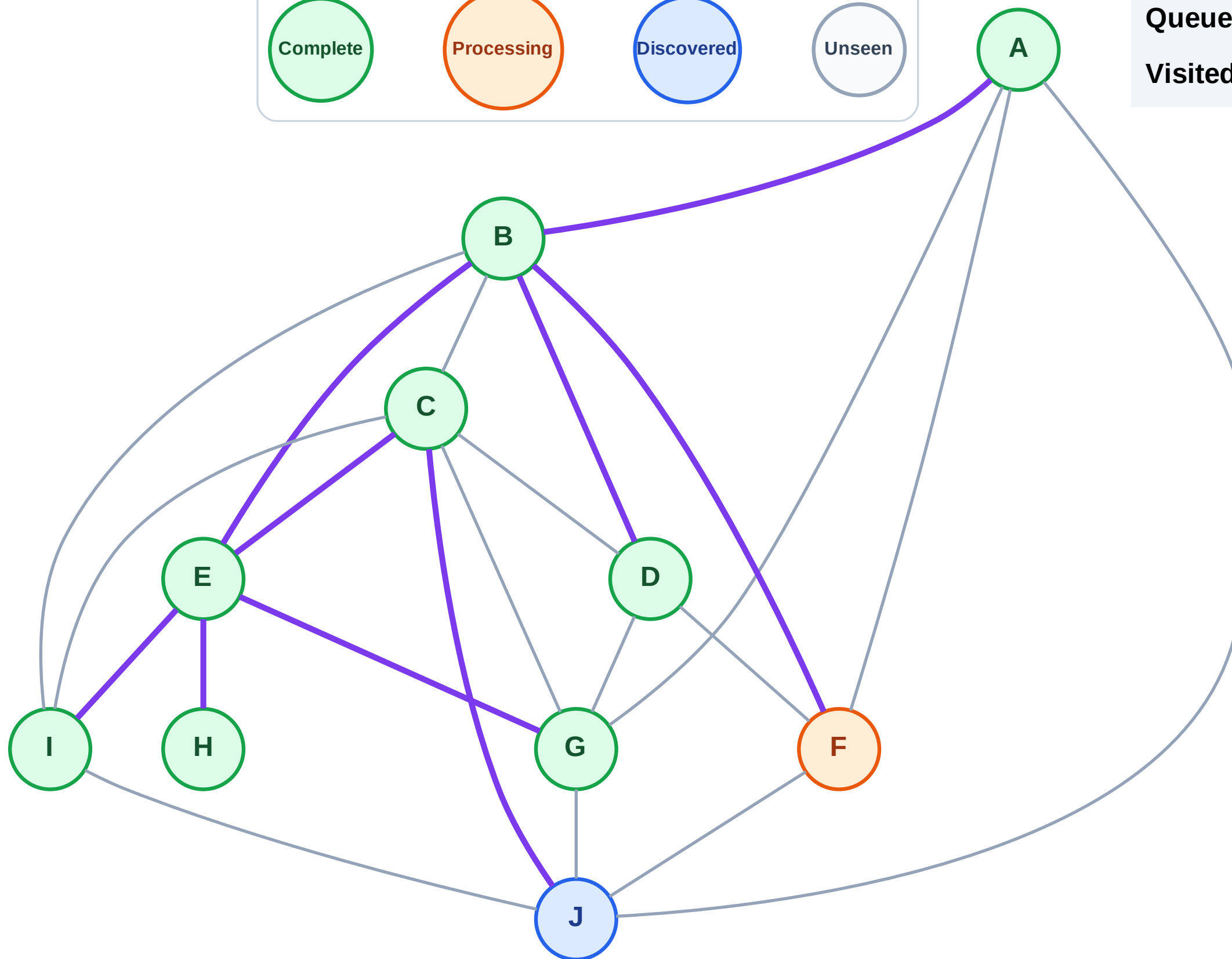
Step 18: Process node F

Legend



Queue: J

Visited: A, B, C, D, E, G, H, I



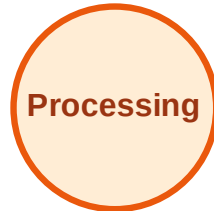
BFS Traversal

Step 19: No new neighbors from F

Legend



Complete



Processing



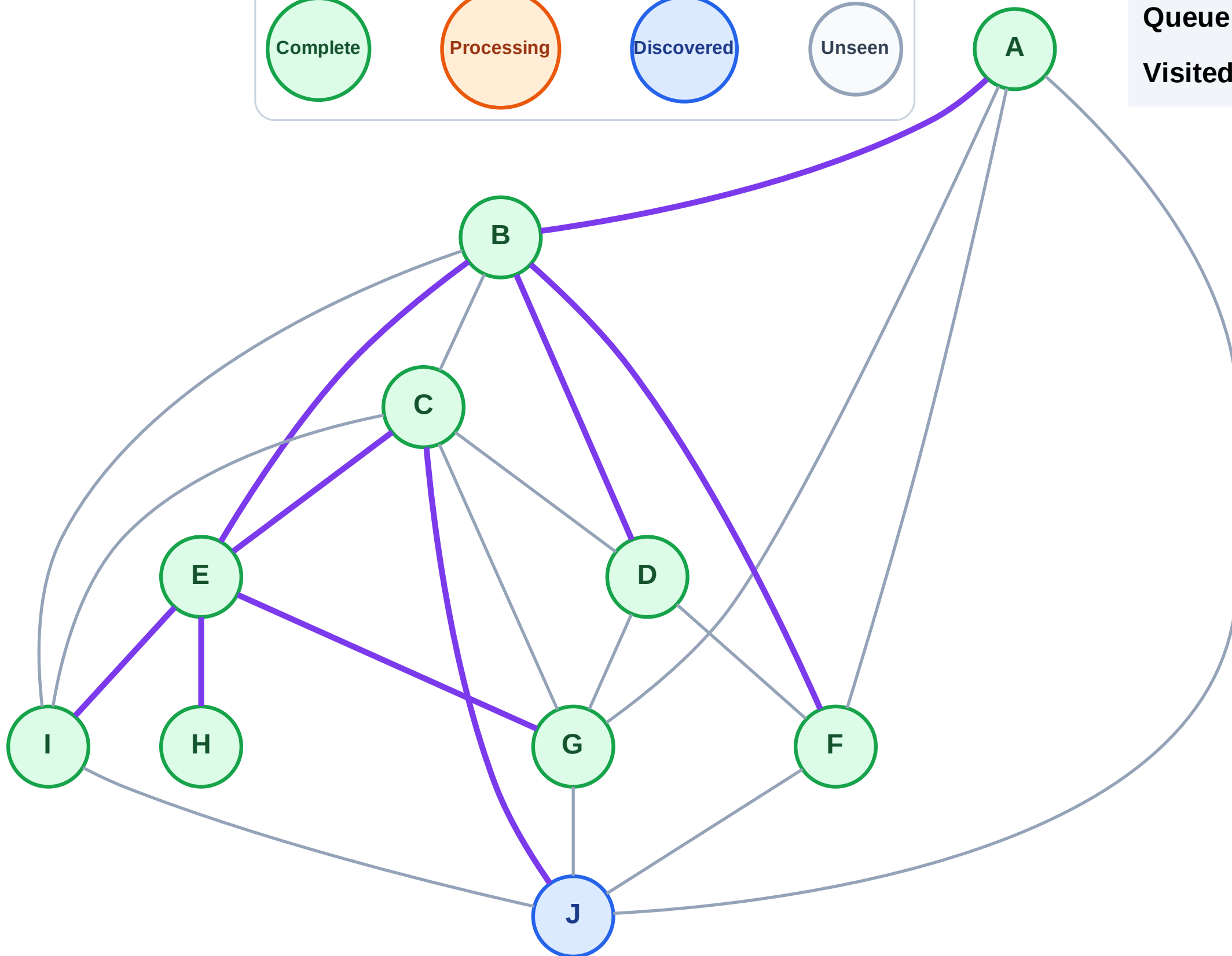
Discovered



Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I



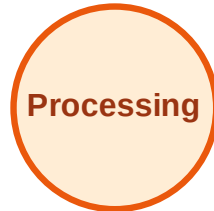
BFS Traversal

Step 20: Process node J

Legend



Complete



Processing



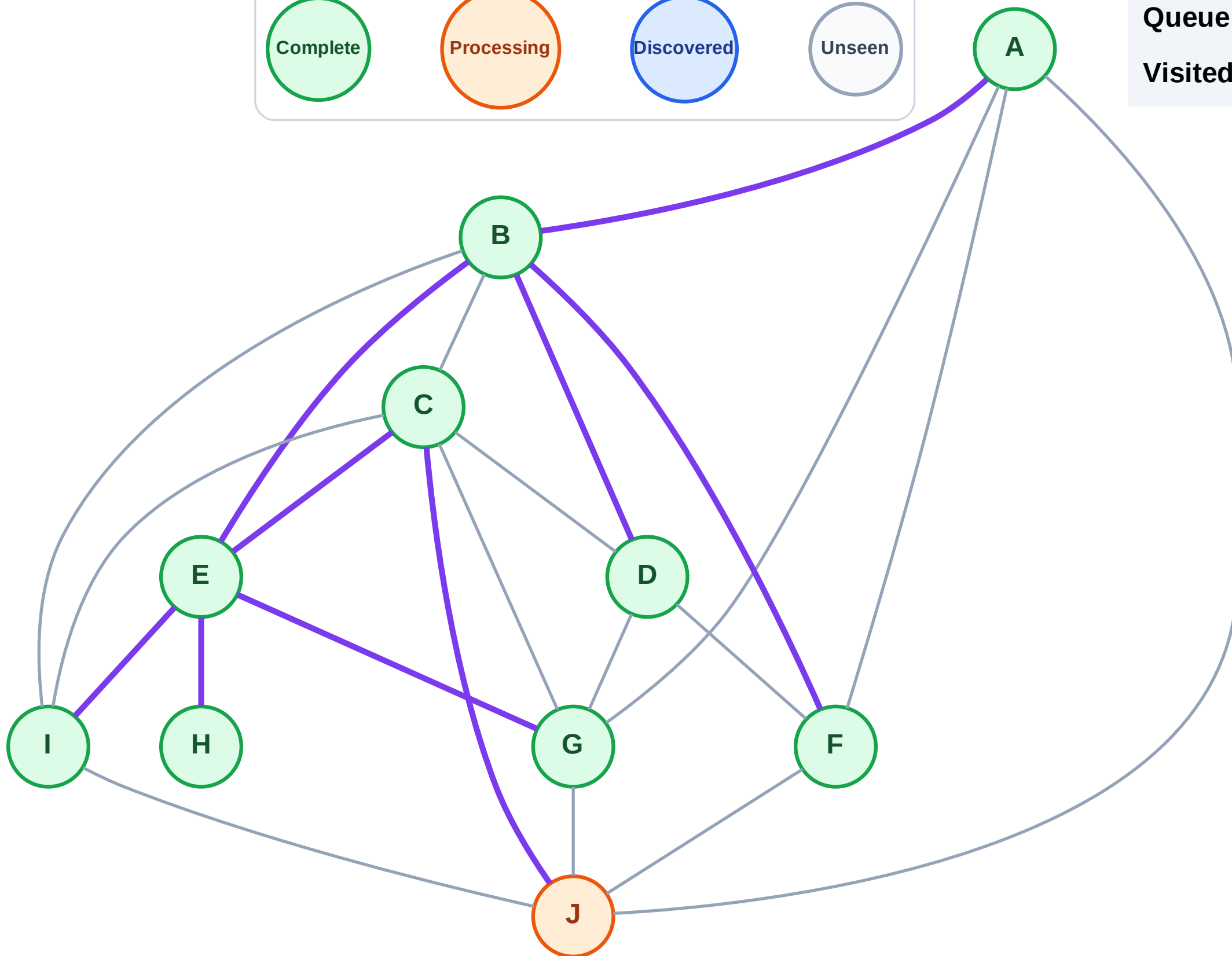
Discovered



Unseen

Queue: (empty)

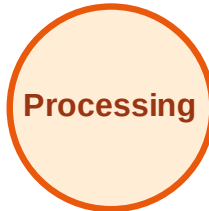
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

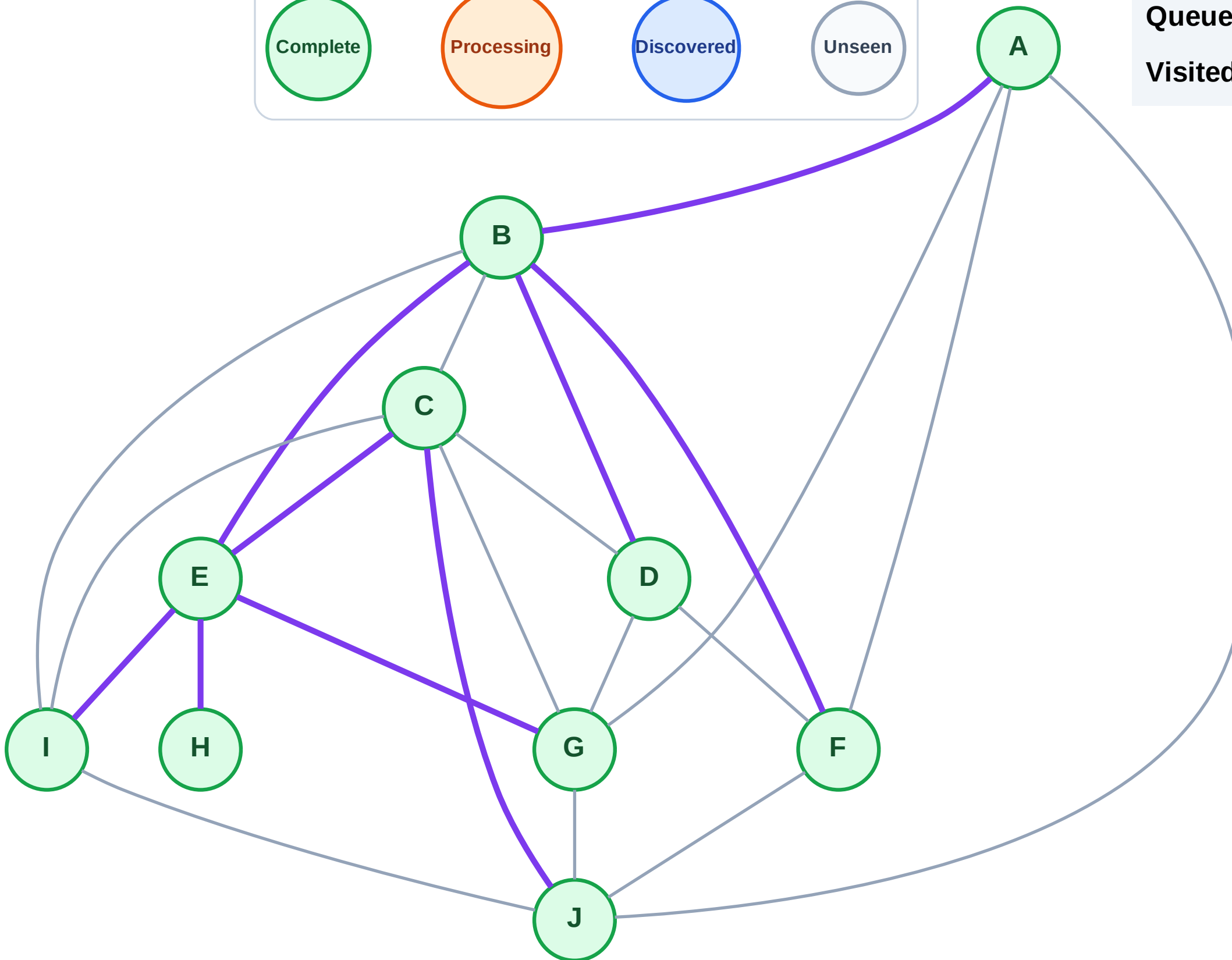
Step 21: No new neighbors from J

Legend



Queue: (empty)

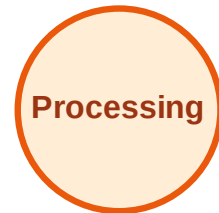
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

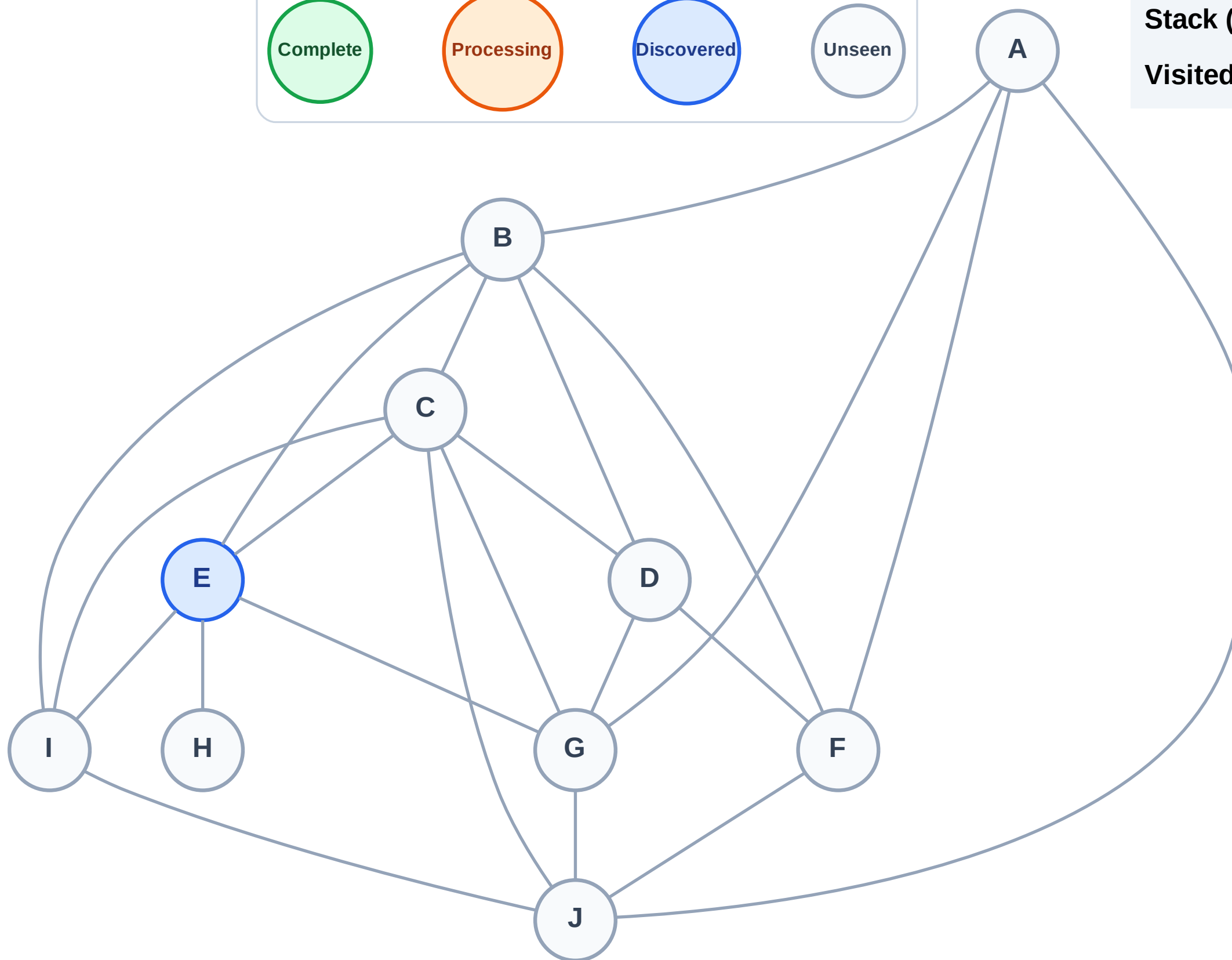
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



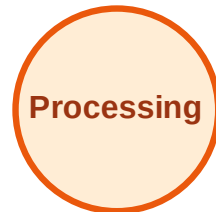
DFS Traversal

Step 2: Process node E

Legend



Complete



Processing



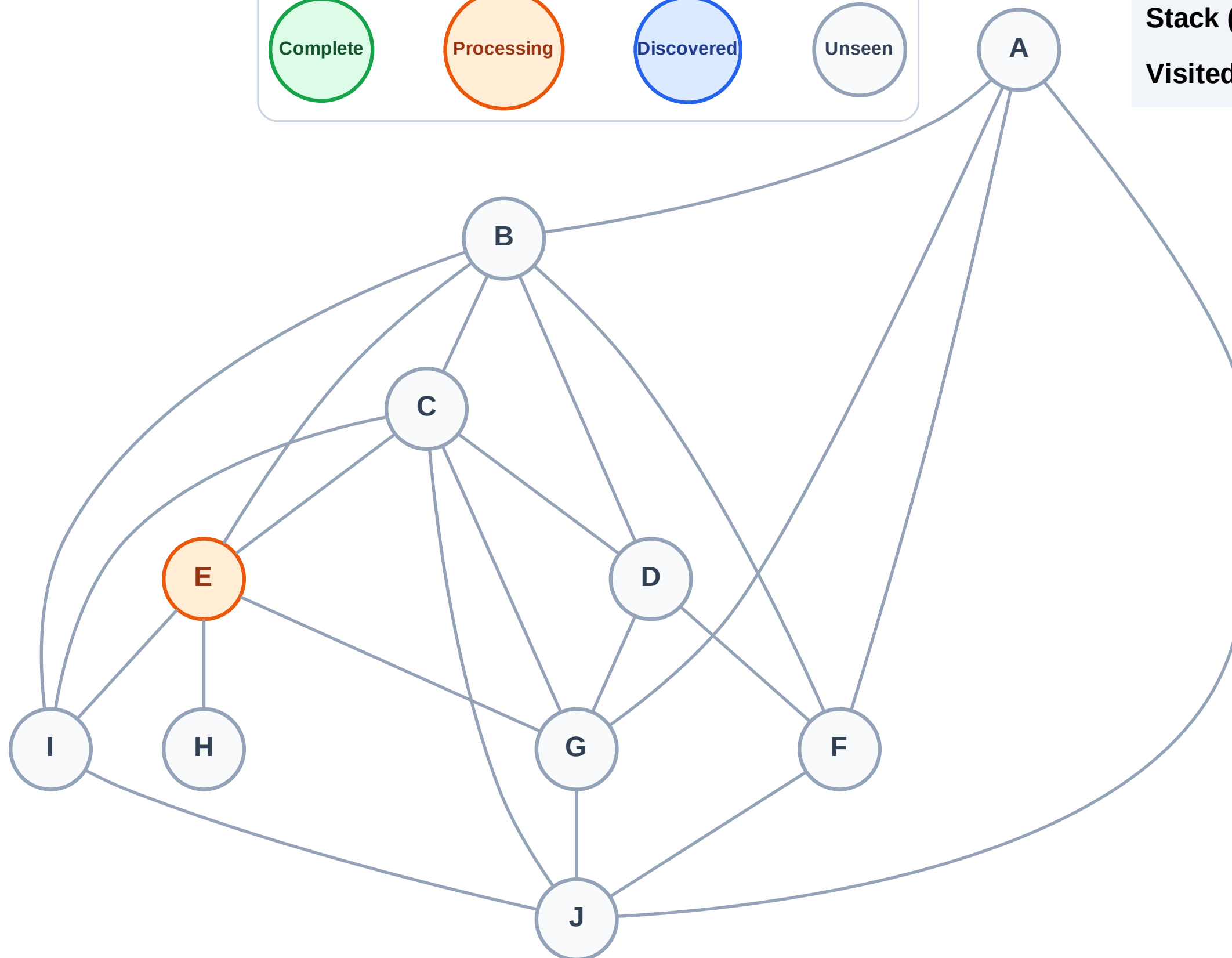
Discovered



Unseen

Stack (top at right): (empty)

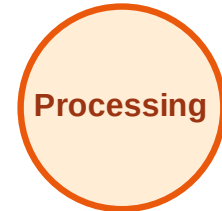
Visited: (none)



DFS Traversal

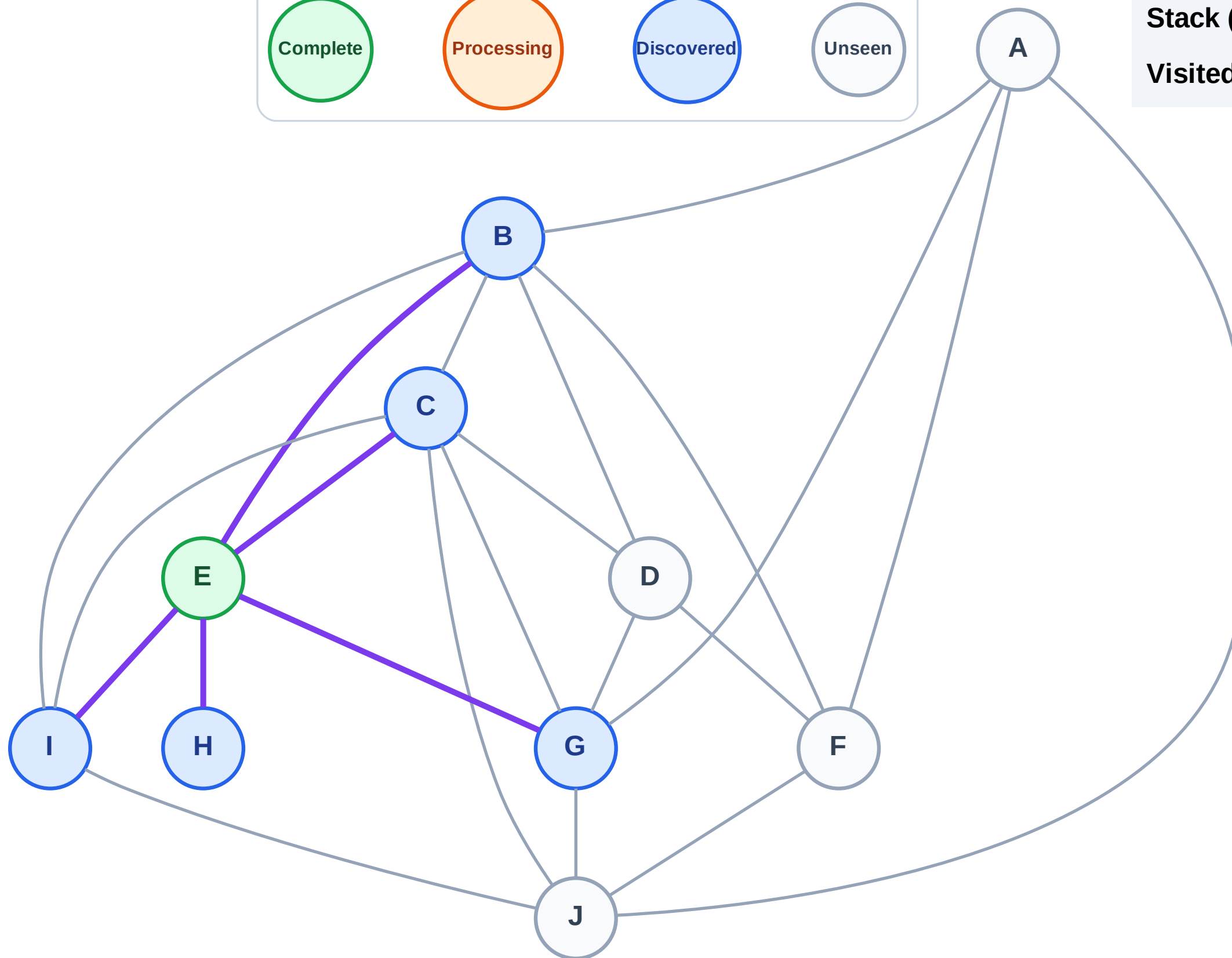
Step 3: Discover I, H, G, C, B from E

Legend



Stack (top at right): I | H | G | C | B

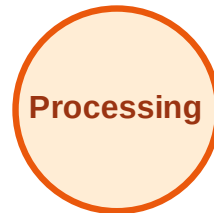
Visited: E



DFS Traversal

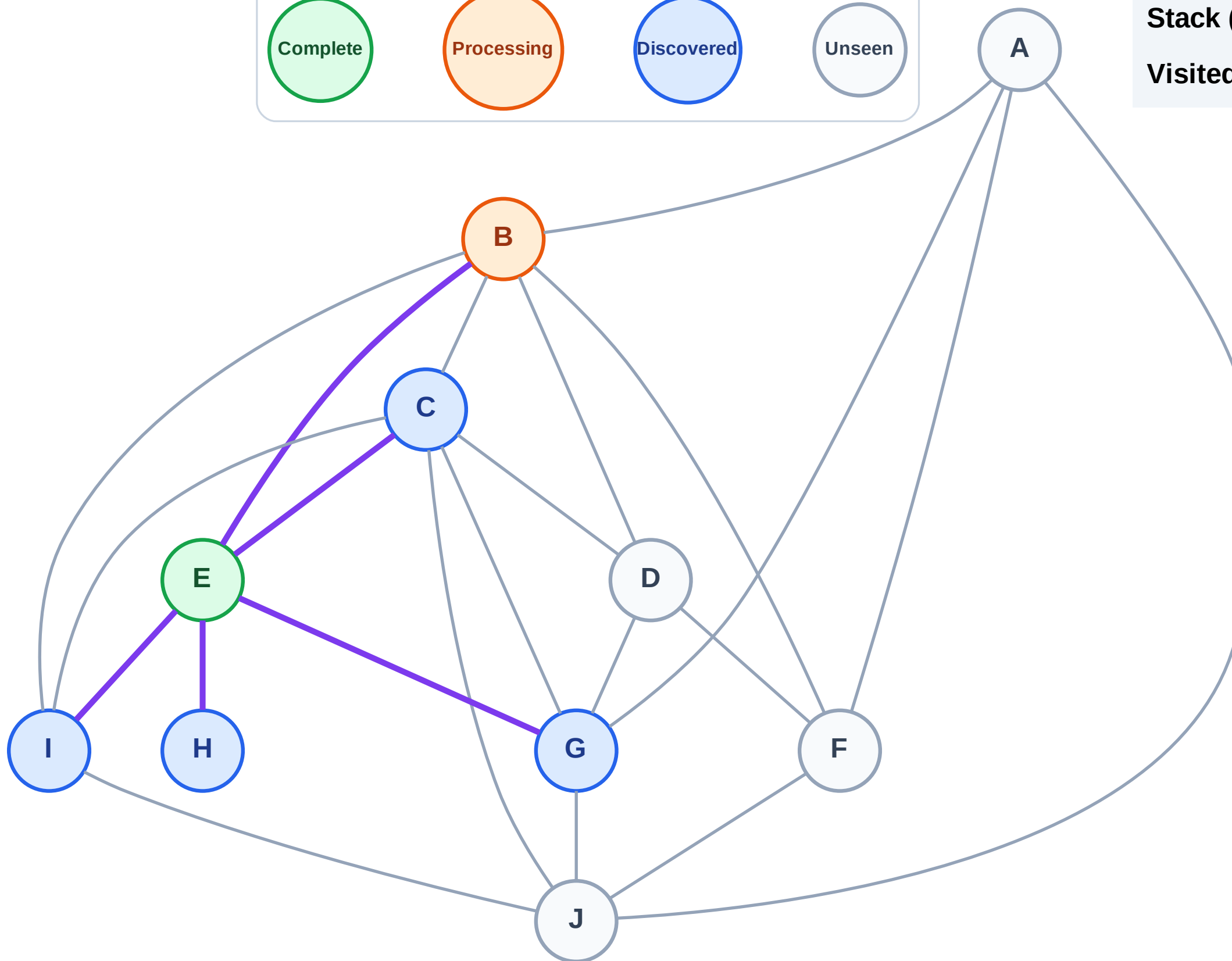
Step 4: Process node B

Legend



Stack (top at right): I | H | G | C

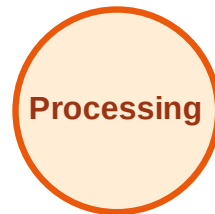
Visited: E



DFS Traversal

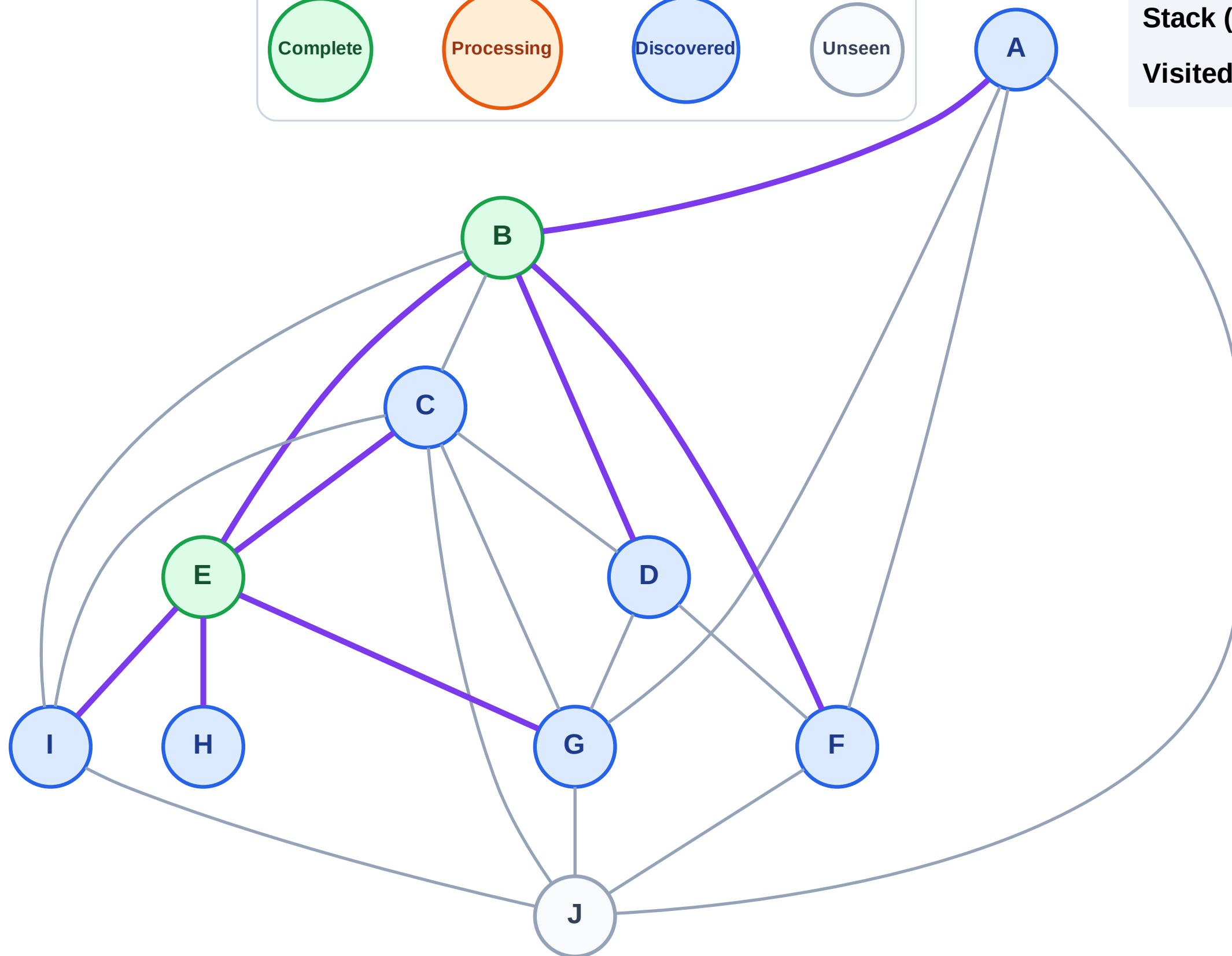
Step 5: Discover F, D, A from B

Legend



Stack (top at right): I | H | G | C | F | D | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

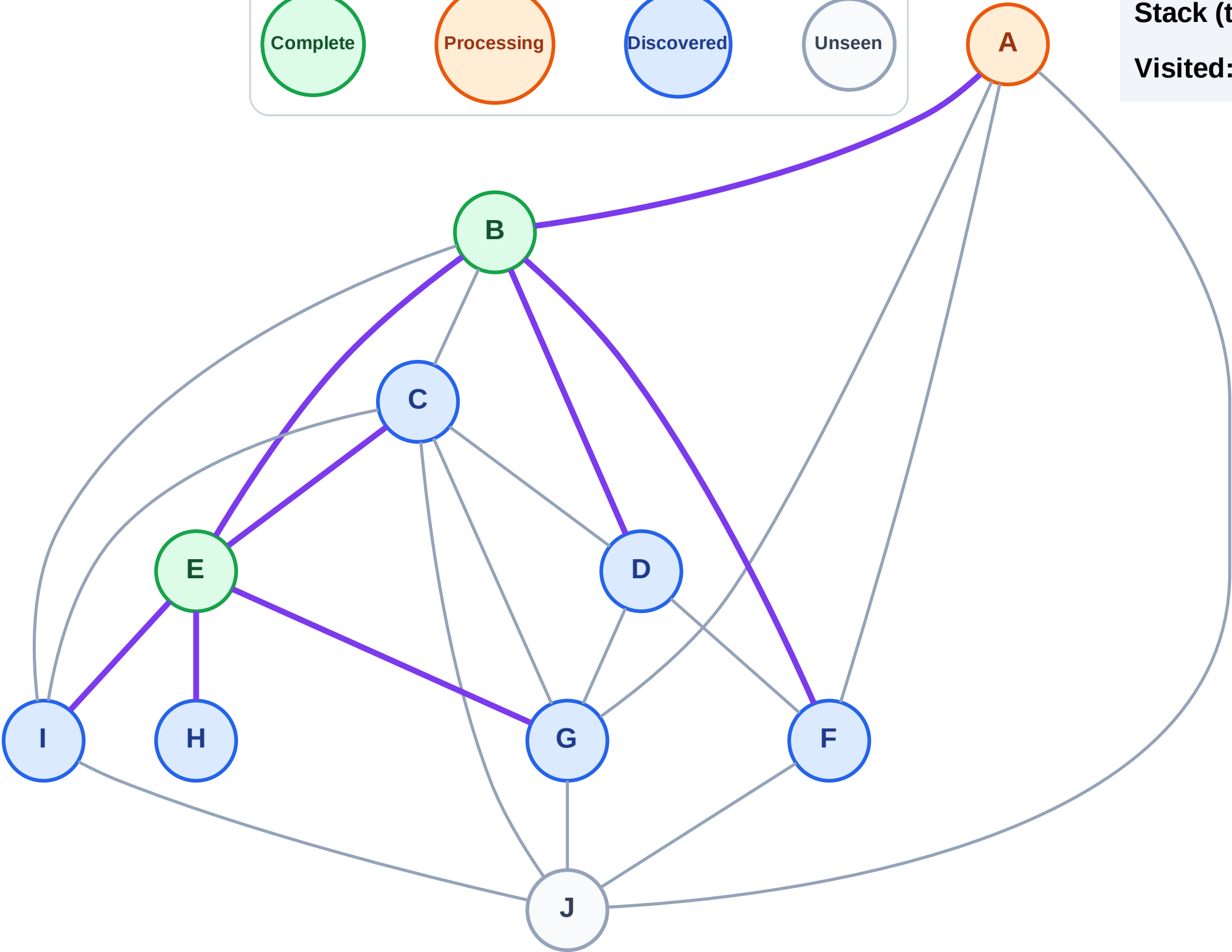
Processing

Discovered

Unseen

Stack (top at right): I | H | G | C | F | D

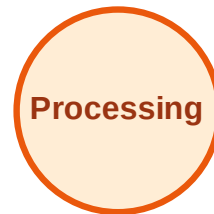
Visited: B, E



DFS Traversal

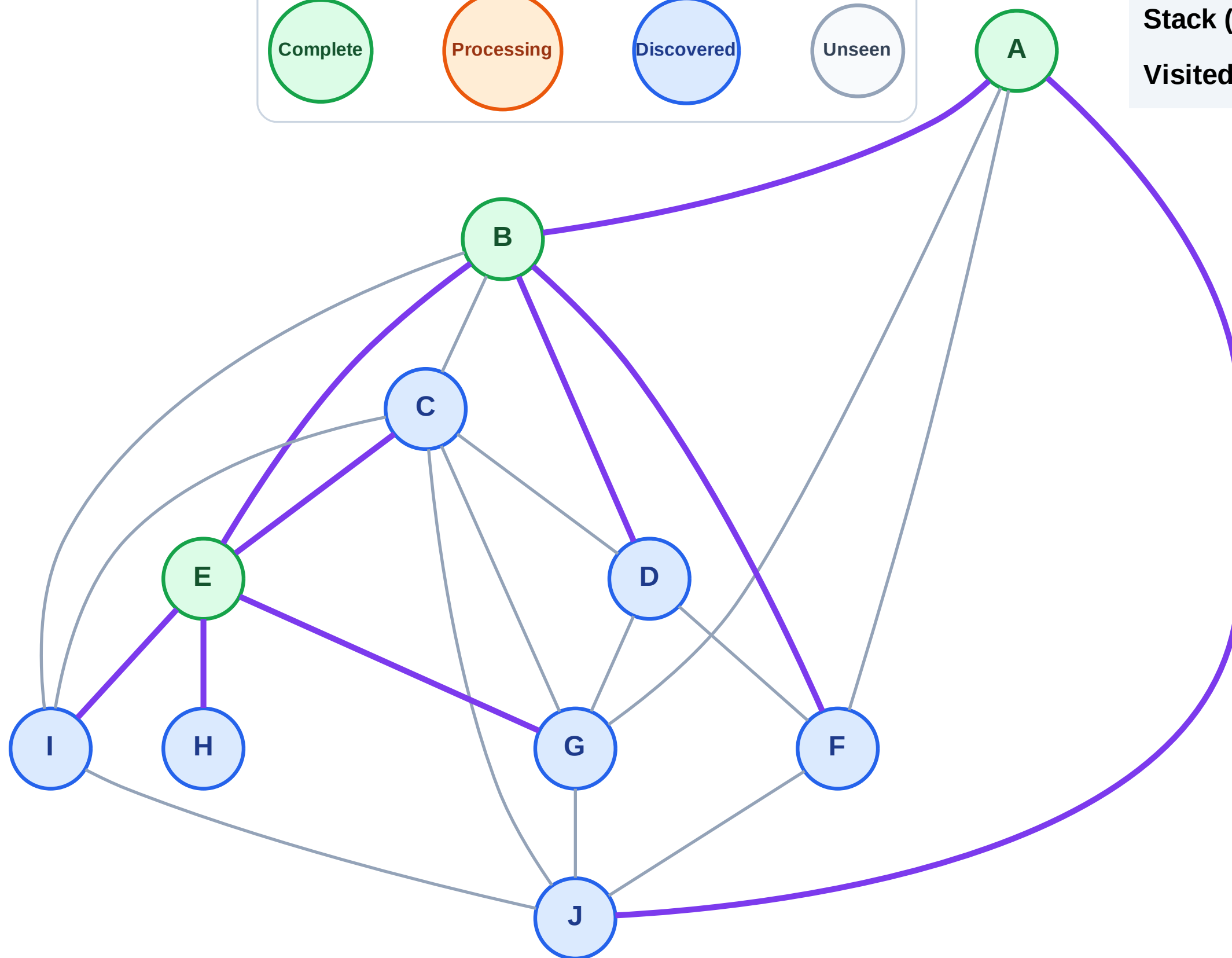
Step 7: Discover J from A

Legend



Stack (top at right): I | H | G | C | F | D | J

Visited: A, B, E



DFS Traversal

Step 8: Process node J

Legend

Complete

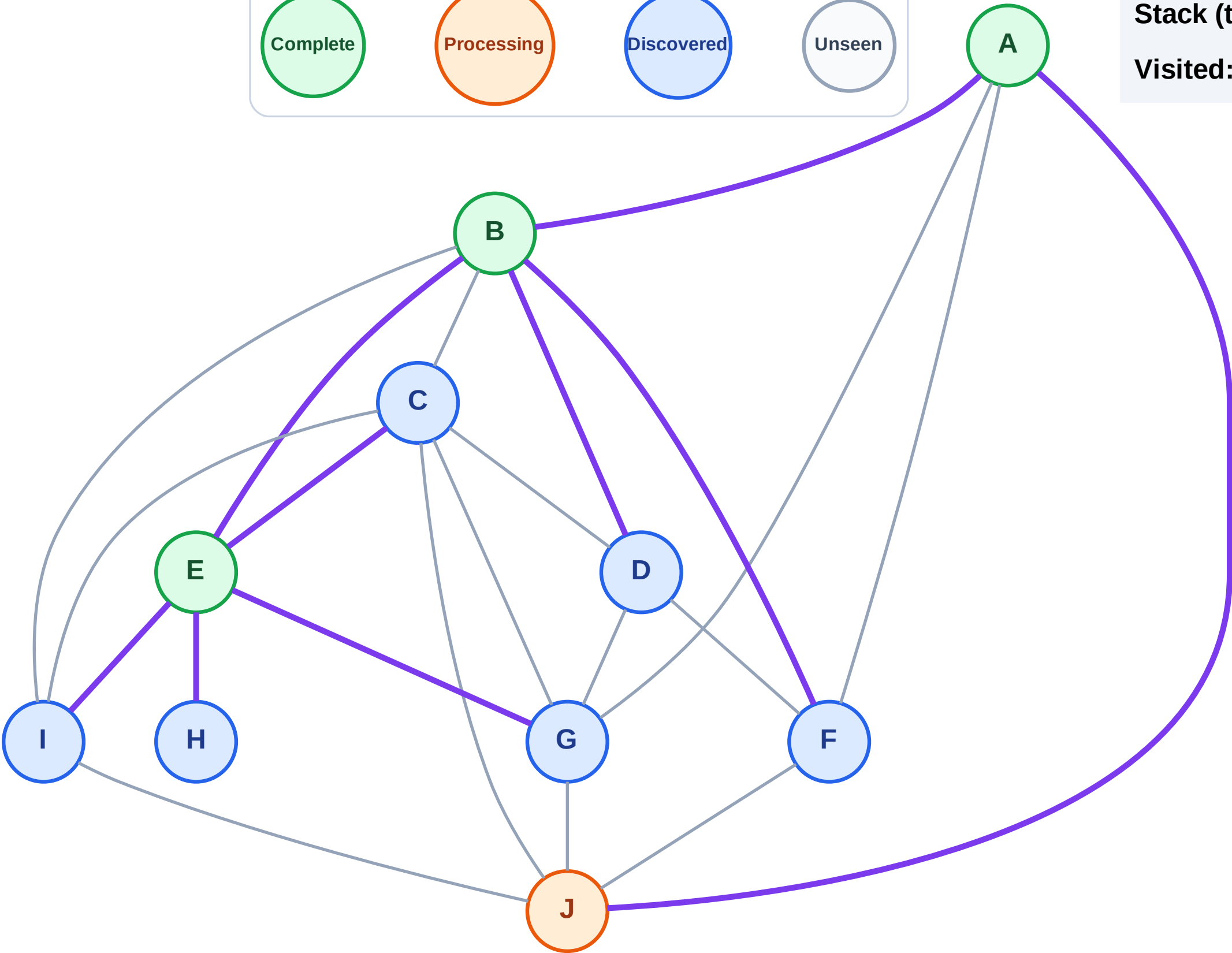
Processing

Discovered

Unseen

Stack (top at right): I | H | G | C | F | D

Visited: A, B, E



Step 9: No new neighbors from J

Legend



The graph illustrates a search process with nodes A through J. The nodes are colored based on their status: Complete (green), Processing (orange), Discovered (blue), and Unseen (grey). The graph shows a network of nodes A through J. Nodes A, B, E, and J are green (Complete). Nodes C, D, F, G, H, and I are blue (Discovered). Nodes I, H, G, and J are also marked with a thick purple line, indicating they are part of the current path or have been visited. The legend at the top right shows the color coding: Complete (green), Processing (orange), Discovered (blue), and Unseen (grey). The legend also includes 'Stack' and 'Visited' labels.

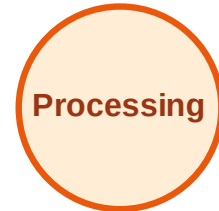
DFS Traversal

Step 10: Process node D

Legend



Complete



Processing



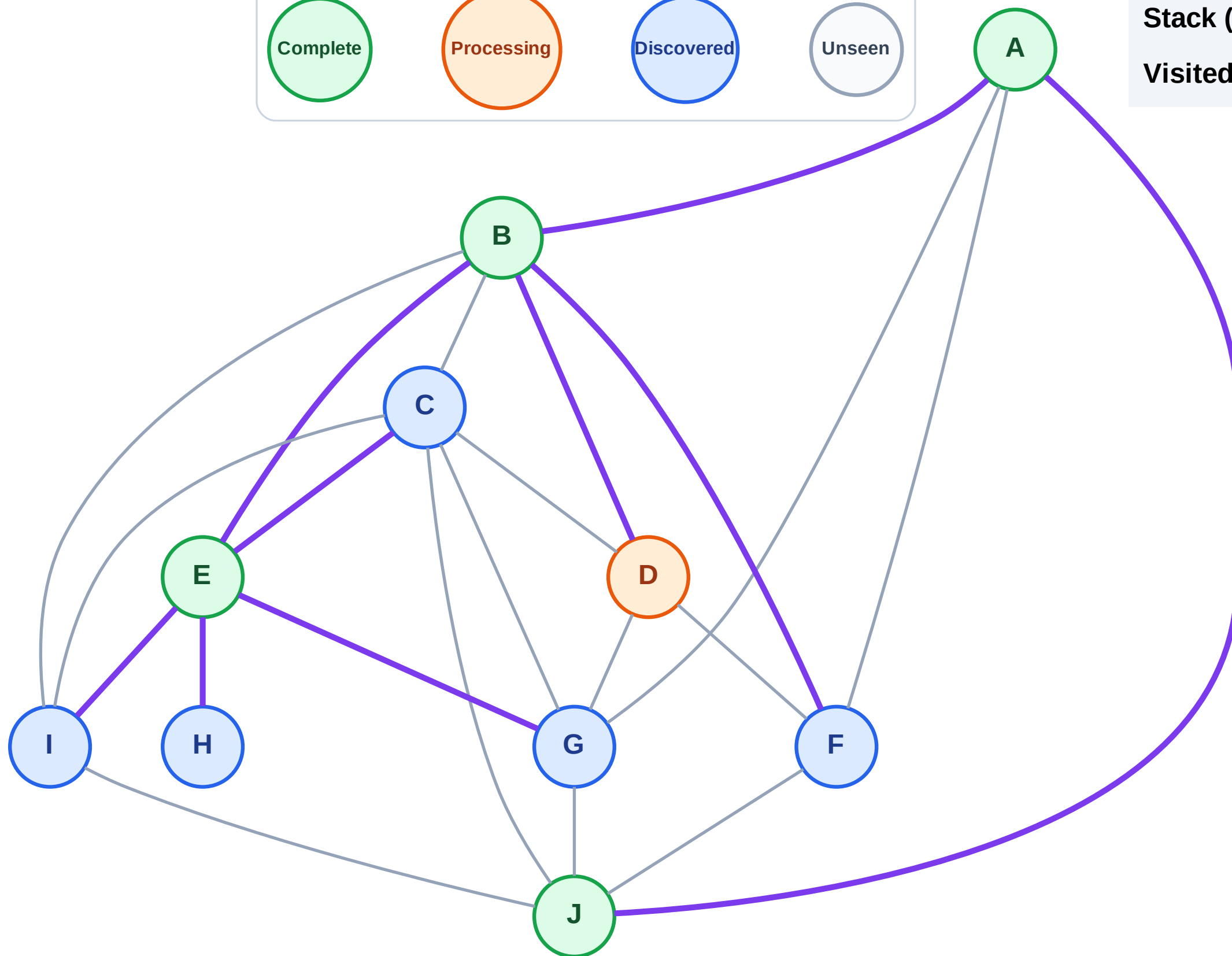
Discovered



Unseen

Stack (top at right): I | H | G | C | F

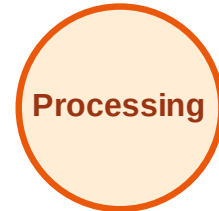
Visited: A, B, E, J



DFS Traversal

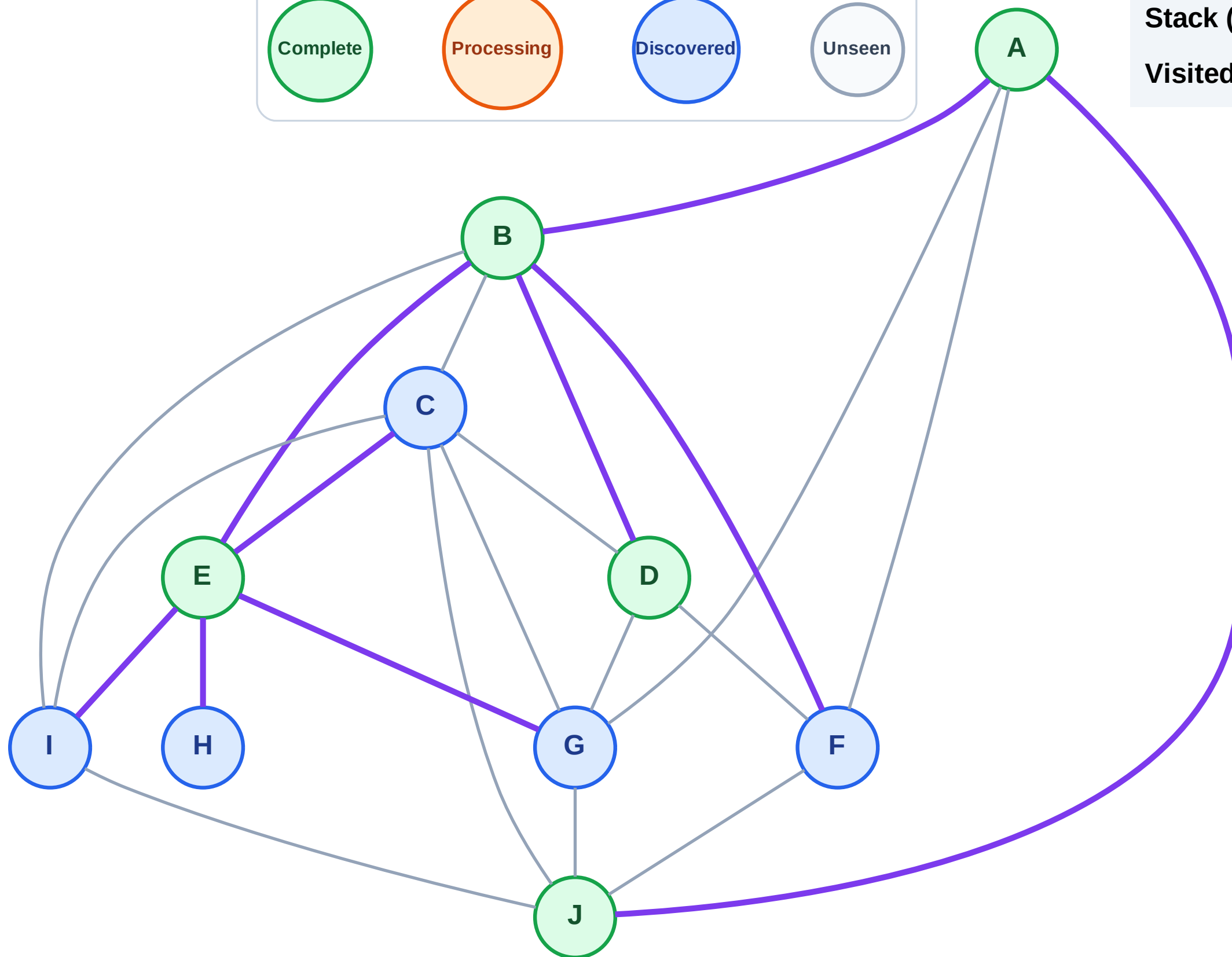
Step 11: No new neighbors from D

Legend



Stack (top at right): I | H | G | C | F

Visited: A, B, D, E, J



DFS Traversal

Step 12: Process node F

Legend

Complete

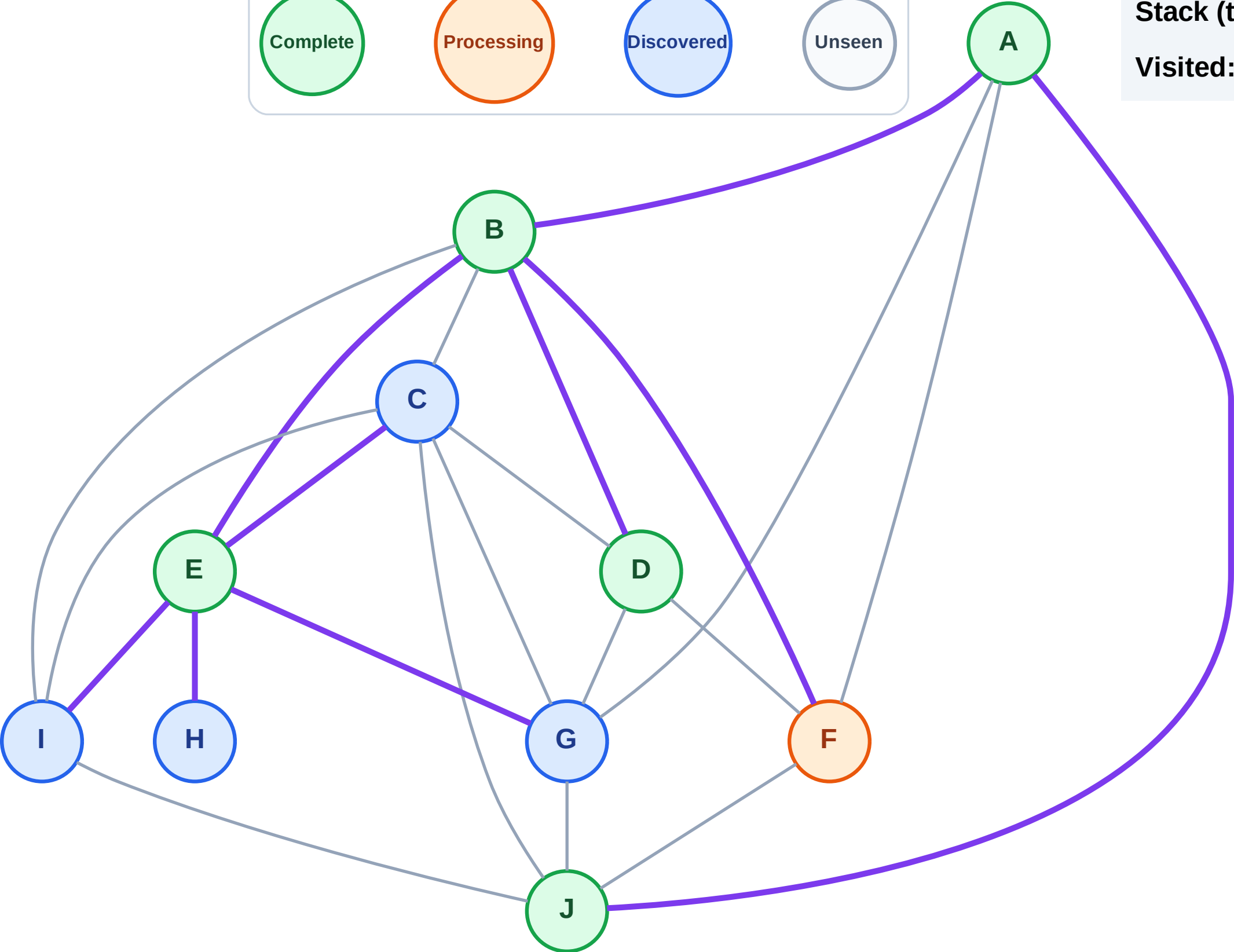
Processing

Discovered

Unseen

Stack (top at right): I | H | G | C

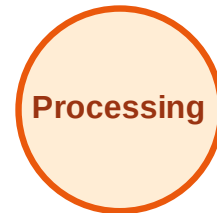
Visited: A, B, D, E, J



DFS Traversal

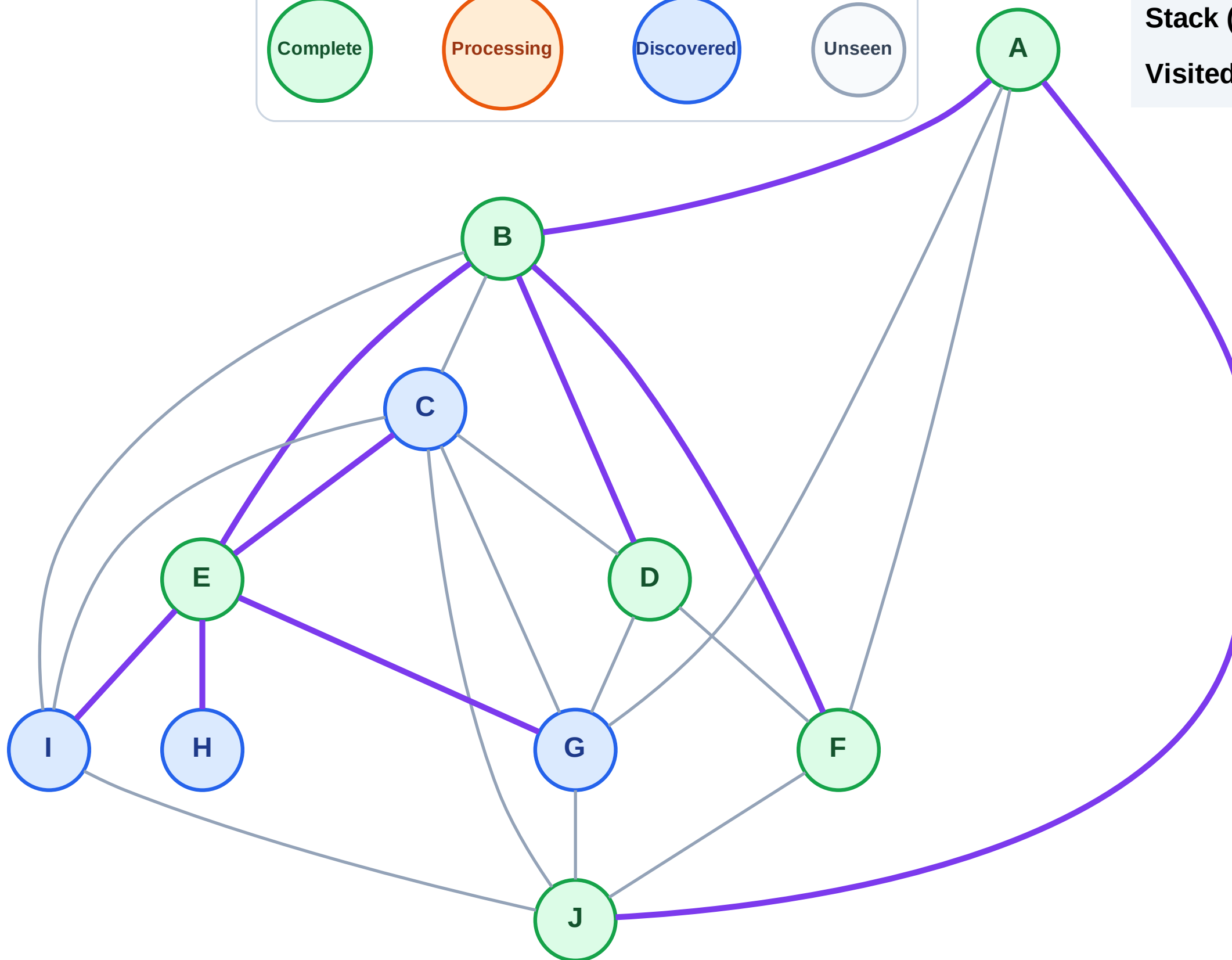
Step 13: No new neighbors from F

Legend



Stack (top at right): I | H | G | C

Visited: A, B, D, E, F, J



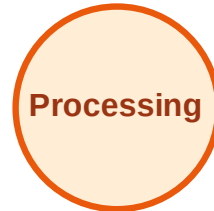
DFS Traversal

Step 14: Process node C

Legend



Complete



Processing



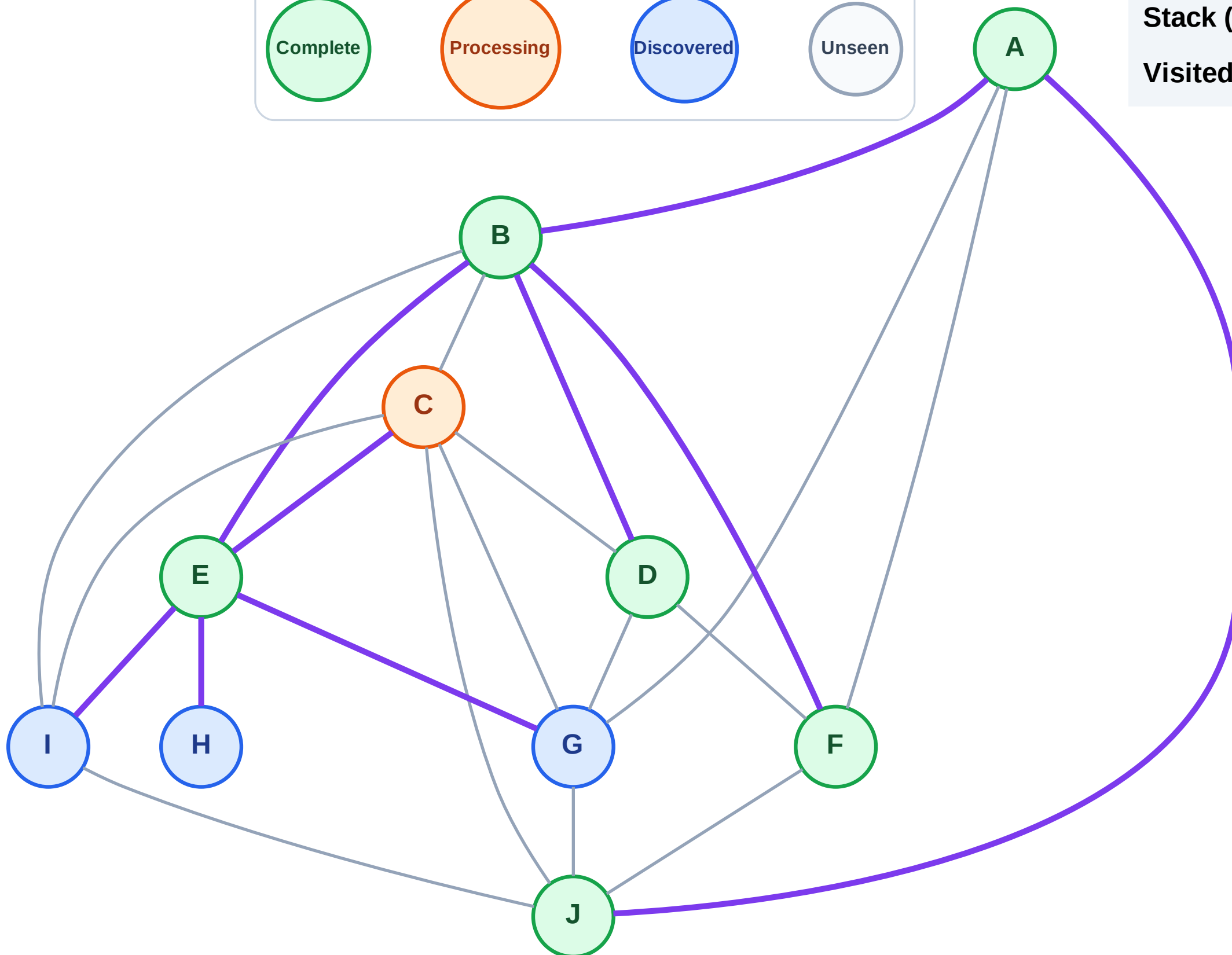
Discovered



Unseen

Stack (top at right): I | H | G

Visited: A, B, D, E, F, J



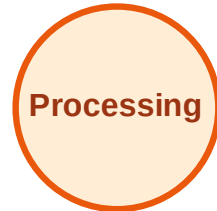
DFS Traversal

Step 15: No new neighbors from C

Legend



Complete



Processing



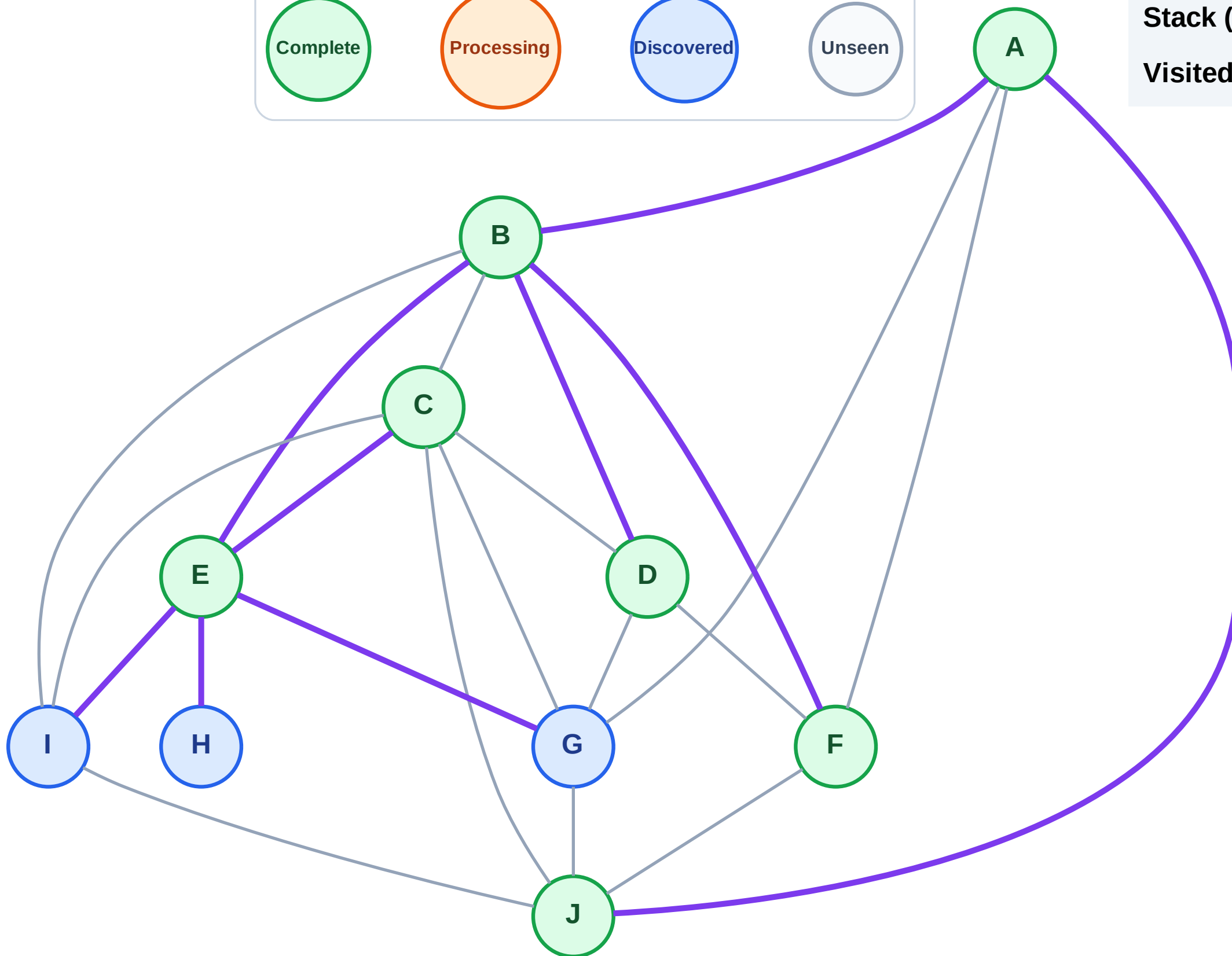
Discovered



Unseen

Stack (top at right): I | H | G

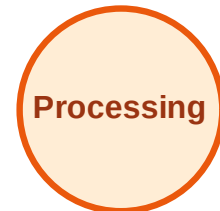
Visited: A, B, C, D, E, F, J



DFS Traversal

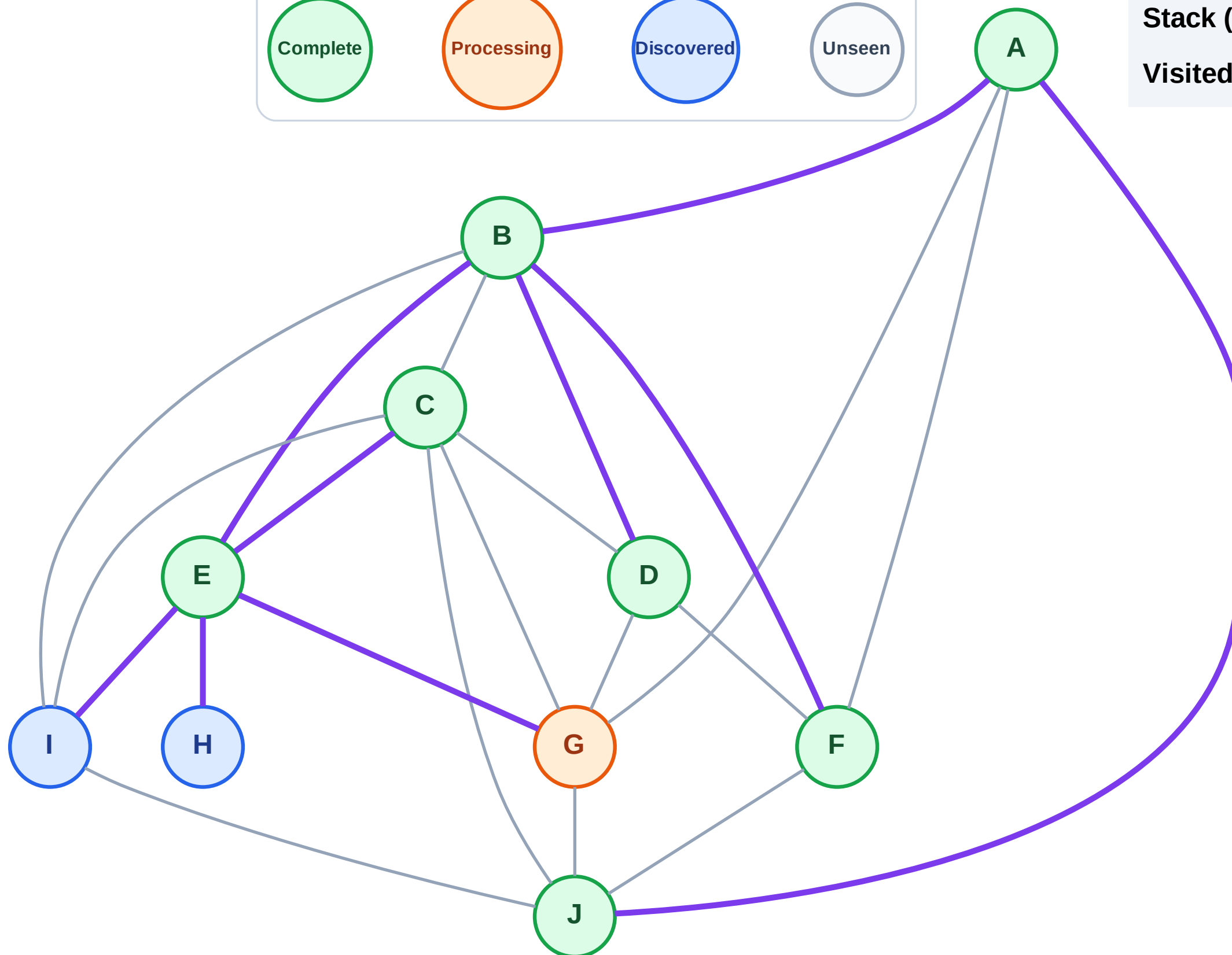
Step 16: Process node G

Legend



Stack (top at right): I | H

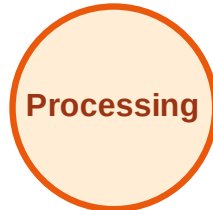
Visited: A, B, C, D, E, F, J



DFS Traversal

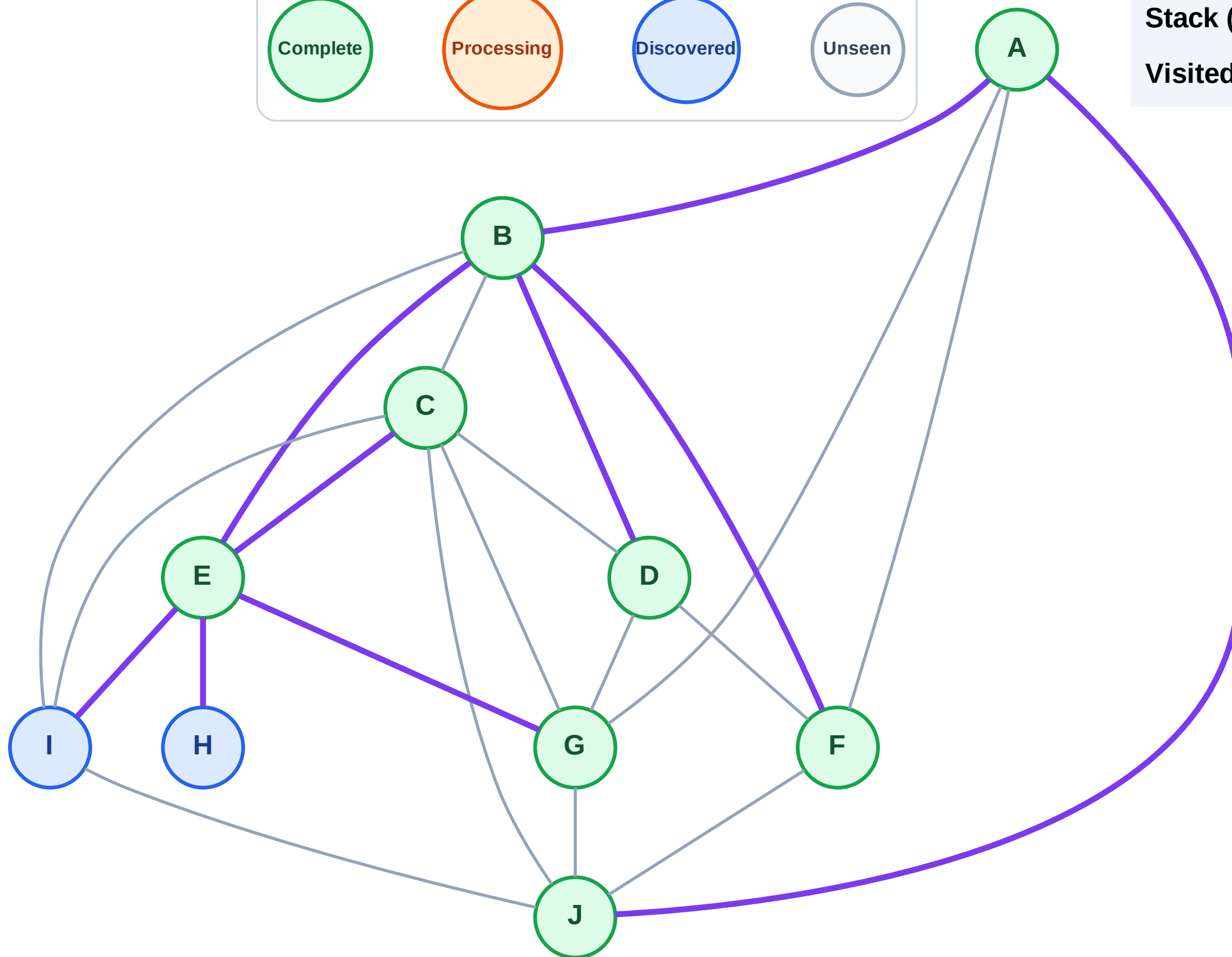
Step 17: No new neighbors from G

Legend



Stack (top at right): I | H

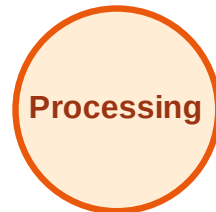
Visited: A, B, C, D, E, F, G, J



DFS Traversal

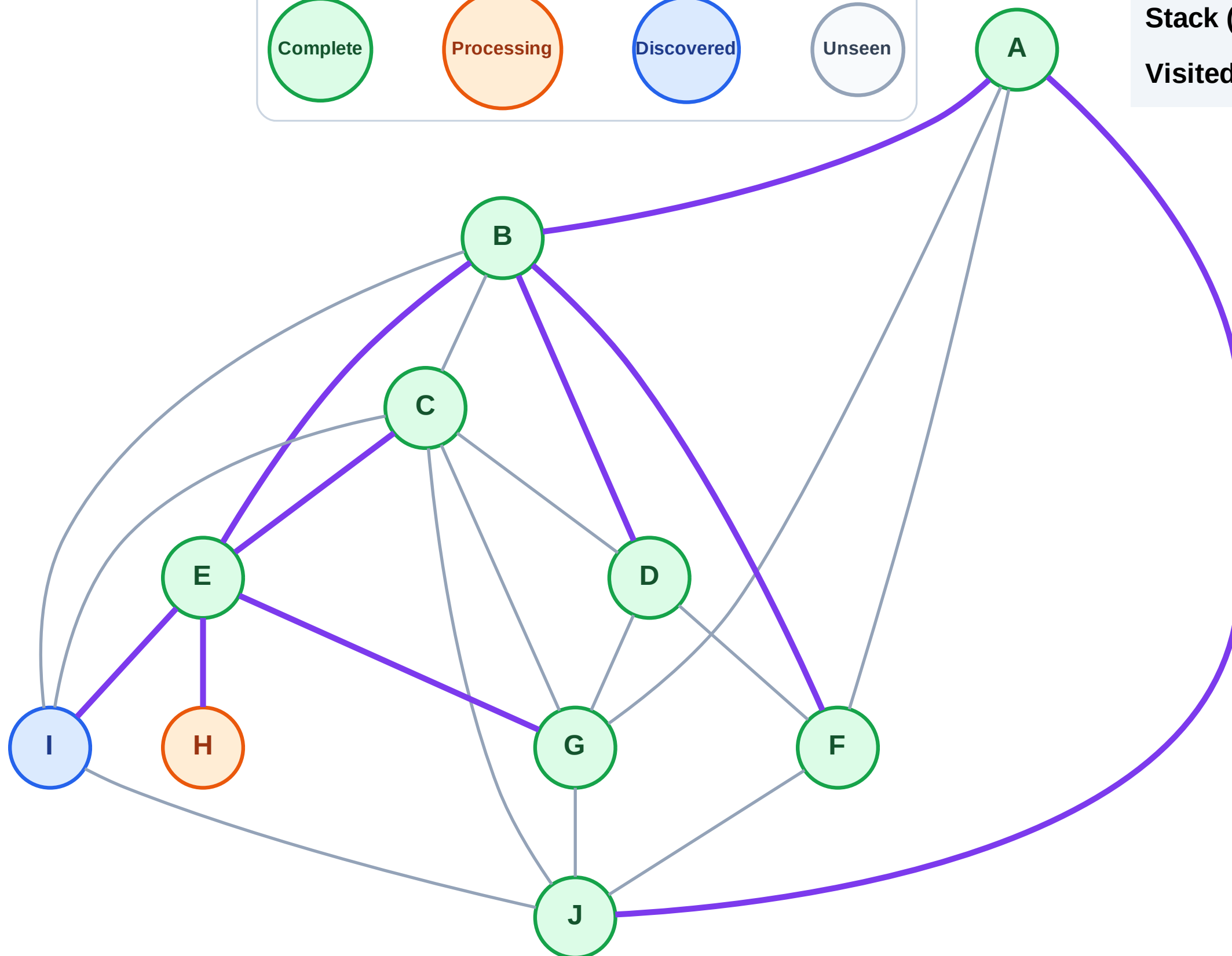
Step 18: Process node H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, J



Step 19: No new neighbors from H

Legend



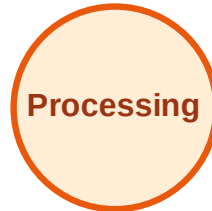
DFS Traversal

Step 20: Process node I

Legend



Complete



Processing



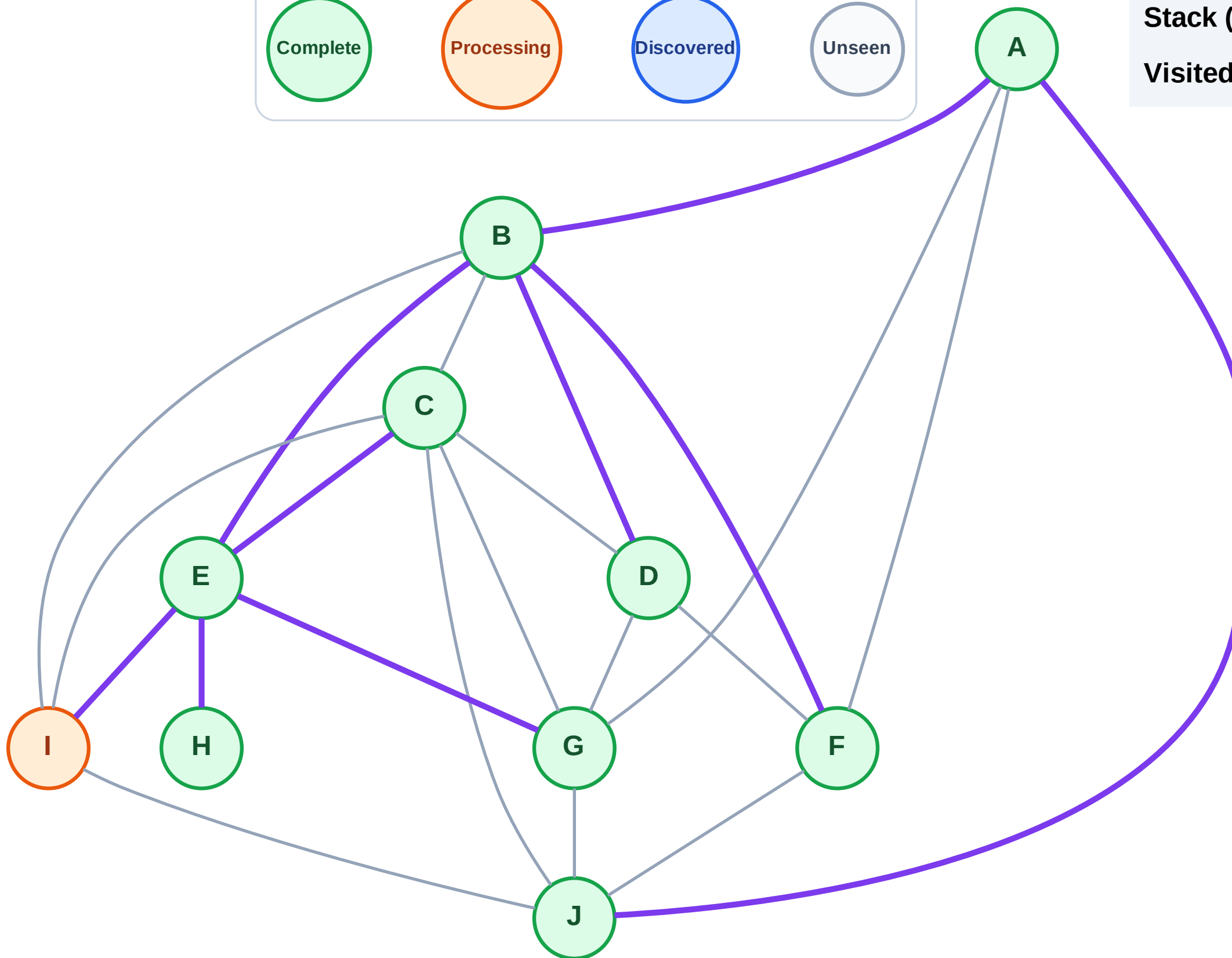
Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

